

Module Outline

- ❑ Thoracic wall
 - Muscles of thoracic wall
 - Intercostal nerve
 - Intercostal vessels
 - Movements of thoracic wall
- ❑ Diaphragm
- ❑ Gross anatomy of respiratory system
- ❑ Development of respiratory system

Anatomy of thoracic wall

By Diliab.D

Introduction

- ❑ **Thorax** is the superior part of trunk between neck and abdomen
 - Contains heart and great vessels, lungs, thymus, trachea and esophagus
 - Organs are constantly moving; **dynamic region**
 - Clinical examination of chest requires a good knowledge of its structure and vital organs it contains

❑ **Function**

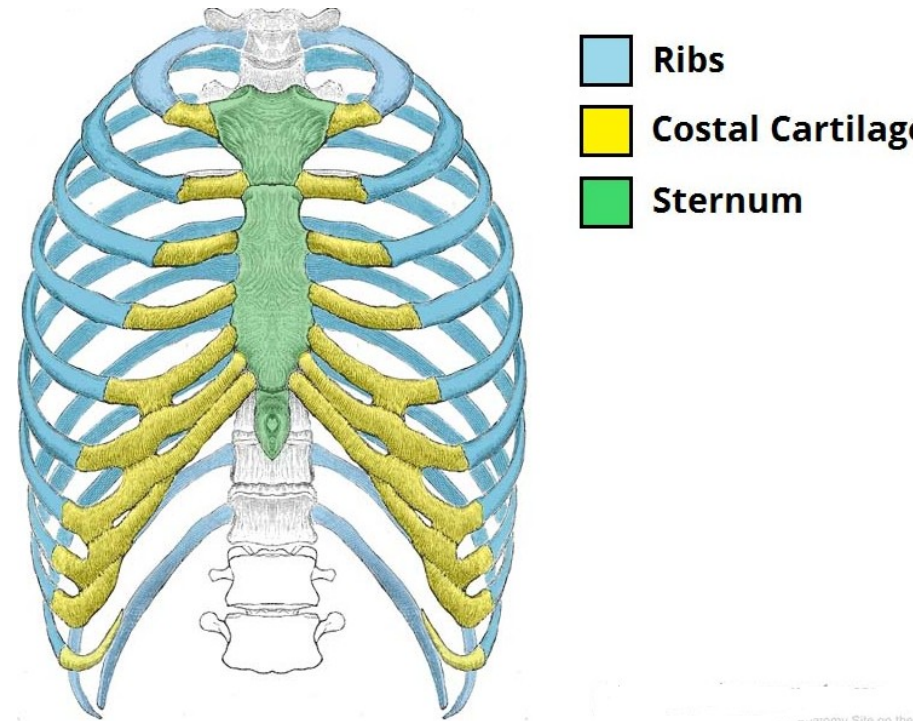
- Provides a **protective** cage around the vital organs (heart, lungs, great blood vessels)
- Provide **attachment** for and **support** the weight of the upper limbs.
- Bony attachment points for muscles of the back, chest and shoulders
- Helps for breathing to happen...etc

Thoracic wall

- ❑ Thoracic wall
 - Muscles of thoracic wall
 - Intercostal nerve
 - Intercostal vessels

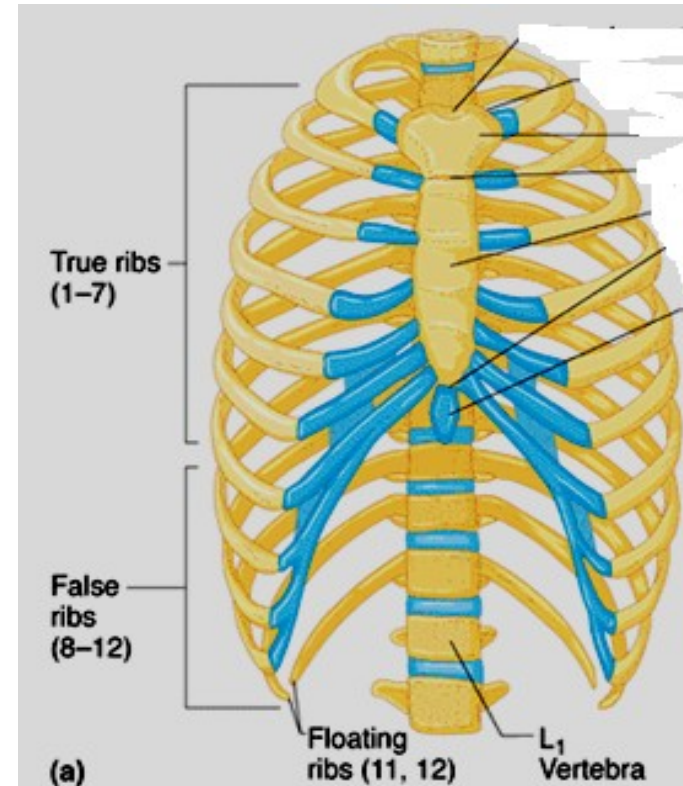
Skeleton of thoracic wall (Thoracic cage)

- **Formed by:** osteocartilaginous thoracic cage
- **Thoracic skeleton includes:**
 - 12 pairs of ribs and costal cartilages
 - 12 thoracic vertebrae and intervertebral discs
 - Sternum

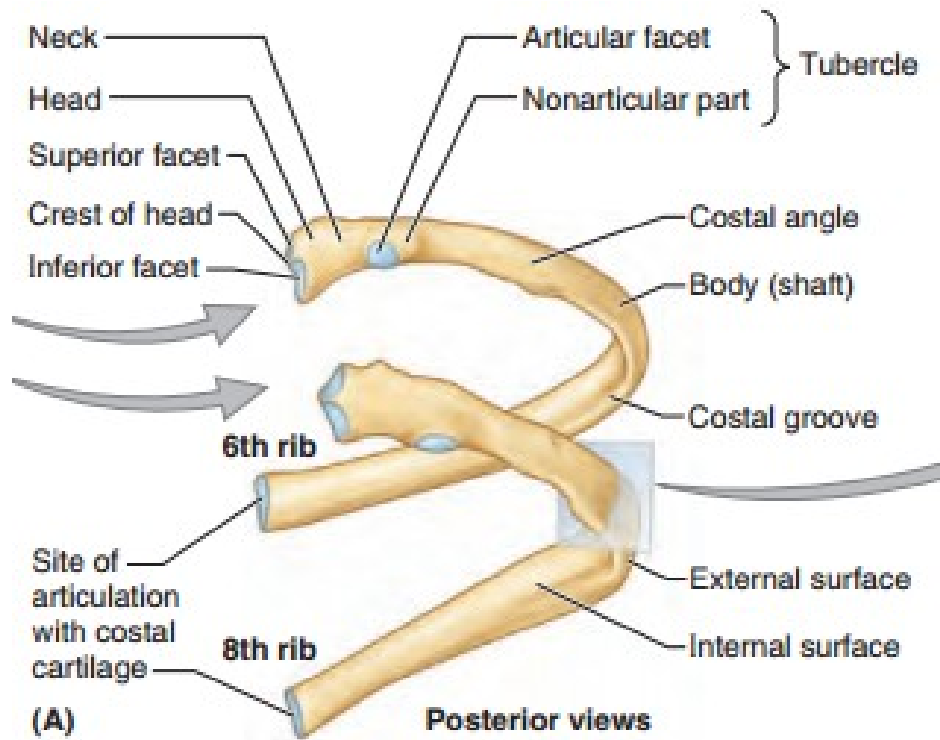


Ribs and costal cartilages

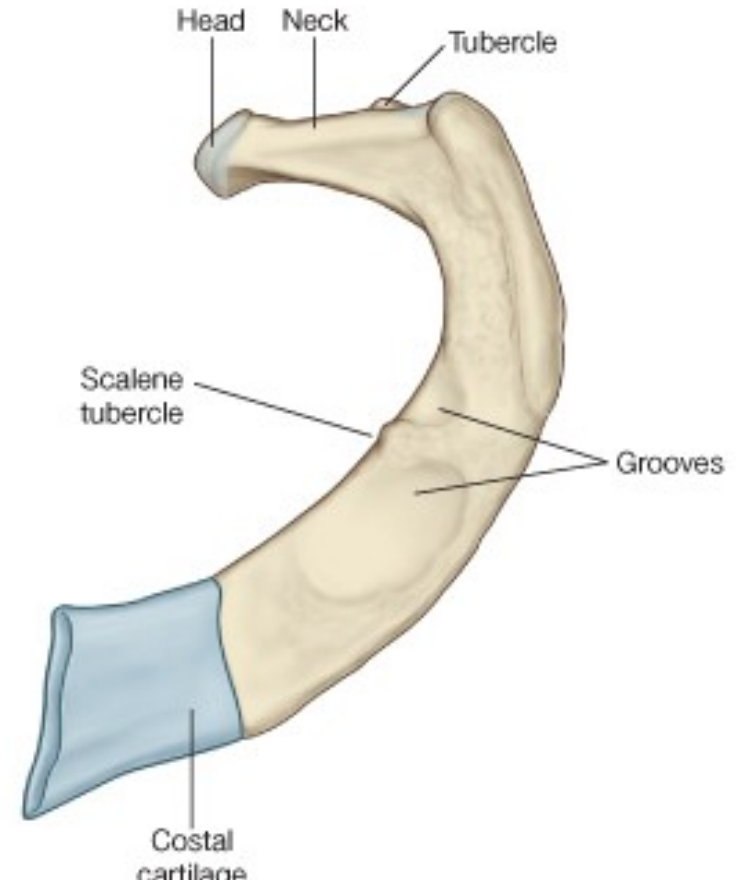
- **Costal cartilages:** prolong the ribs anteriorly and contribute to the **elasticity** of the thoracic wall.
- **Ribs** are curved, flat bones that form most of the thoracic cage
 - **Twelve pairs**
 - ❑ Based on their attachment with sternum there are three types of ribs
 - **True ribs (vertebrosternal):** ribs 1-7 attach directly to sternum by separate costal cartilages
 - **False ribs (vertebrochondral):** ribs 8-10 attach indirectly to sternum by attaching to costal cartilages immediately above
 - **Floating (vertebral, free) ribs:** ribs 11-12 have no anterior attachments



- Based on there **structural features** ribs can be divided into
 - Typical ribs (**3rd-9th**)
 - Atypical ribs (**1st, 2nd, & 10th-12th**)



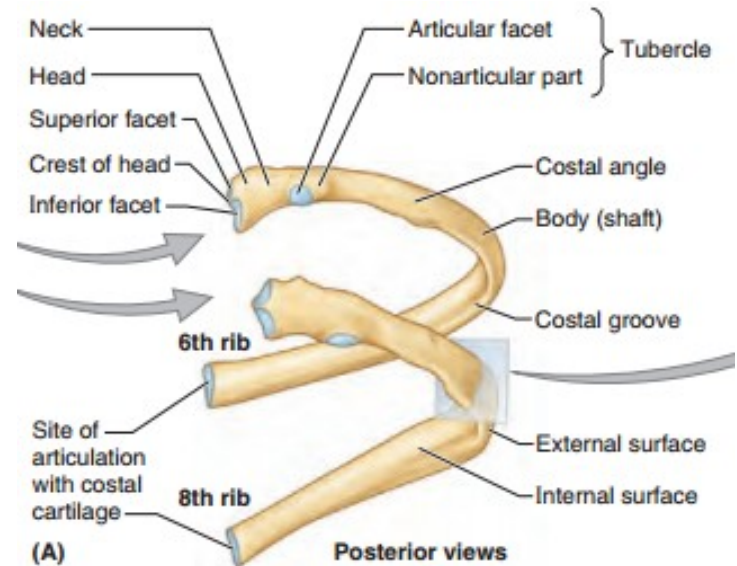
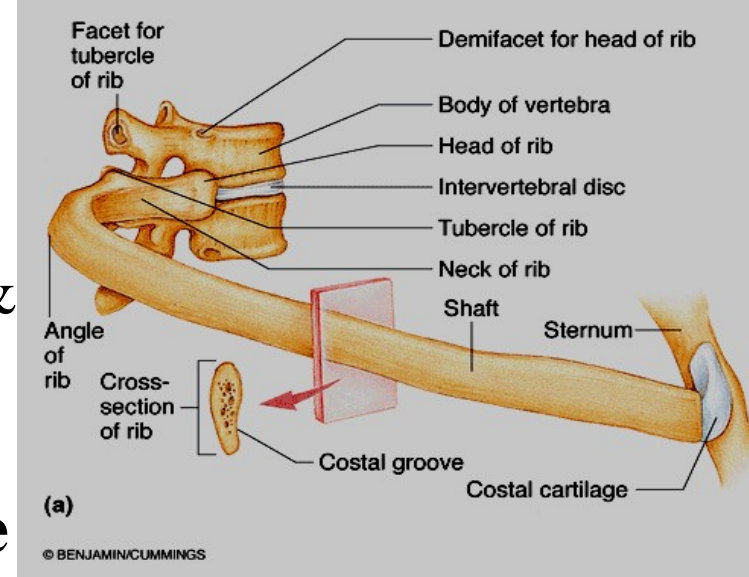
Rib I



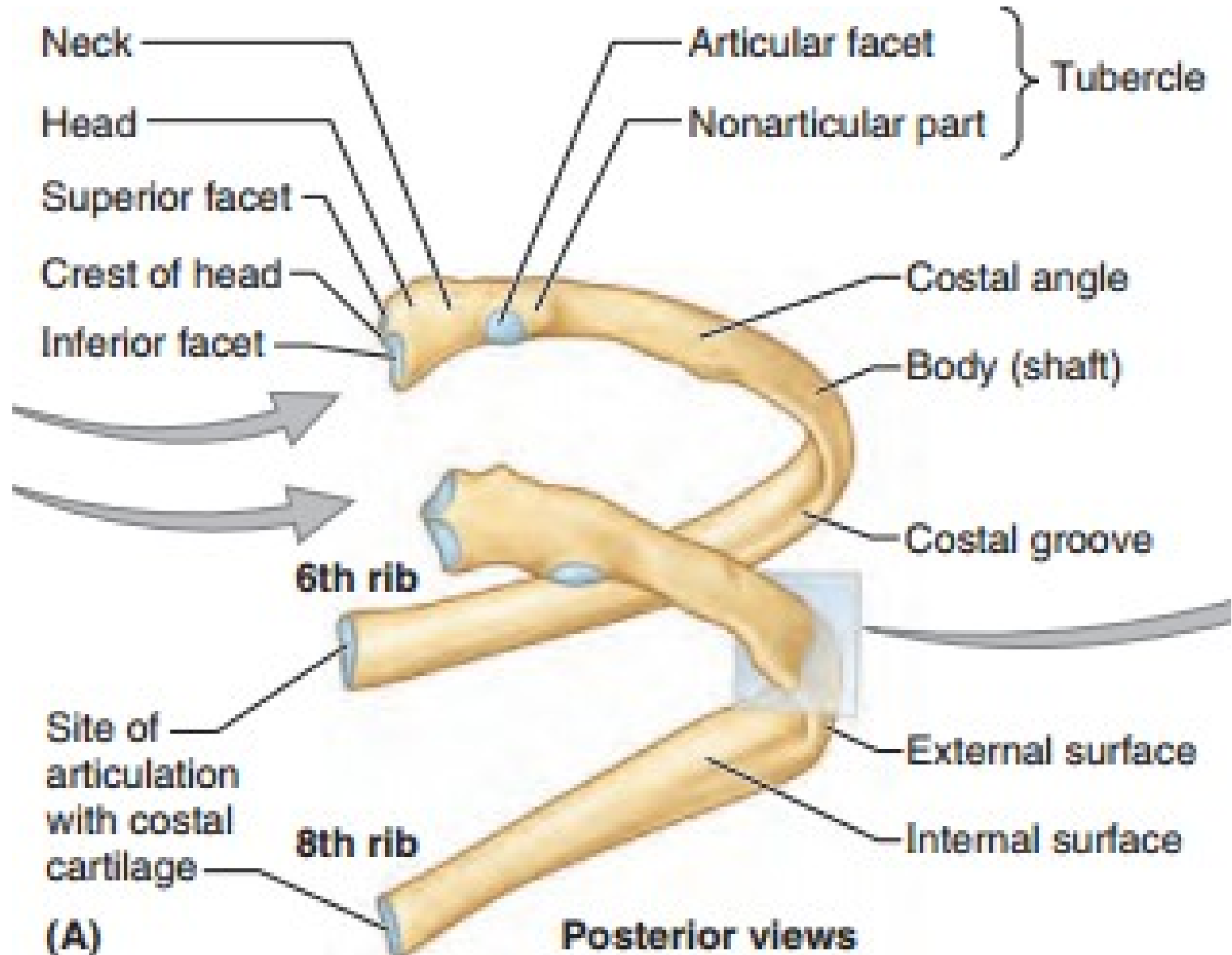
Typical ribs (3rd-9th)

□ Common features of typical ribs

- **Head:** has 2 facets
 - One facet to articulate with the numerically corresponding vertebra &
 - The other facet for the vertebra superior to it.
- **Tubercle:** articulates with **transverse process** of corresponding vertebra
- **Neck:** between head and tubercle
- **Shaft (body):** flat, curved at costal angle where the rib turns anterolaterally.
- **Costal groove:** protect intercostal nerve and vessels



Typical ribs



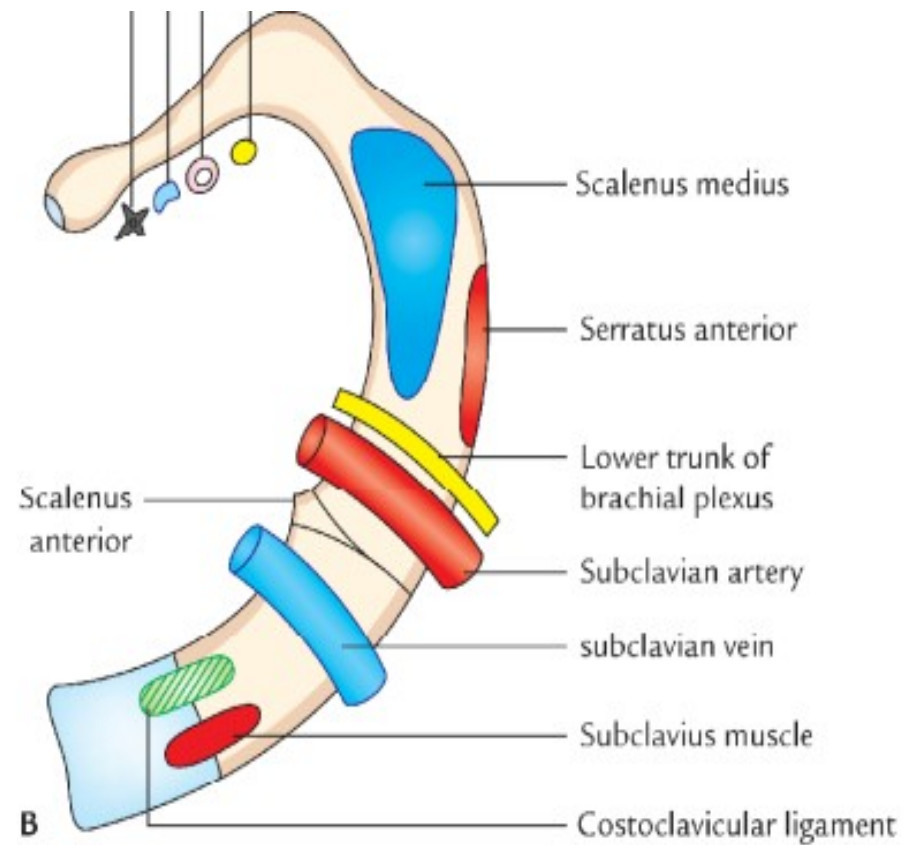
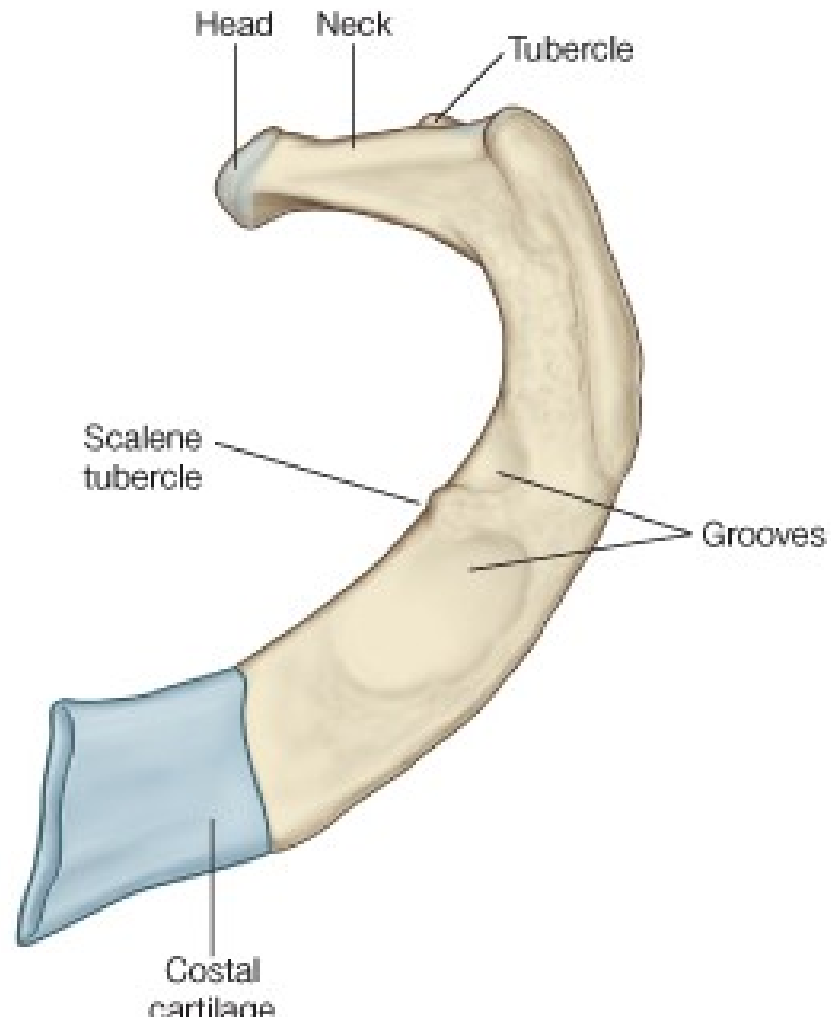
Atypical ribs

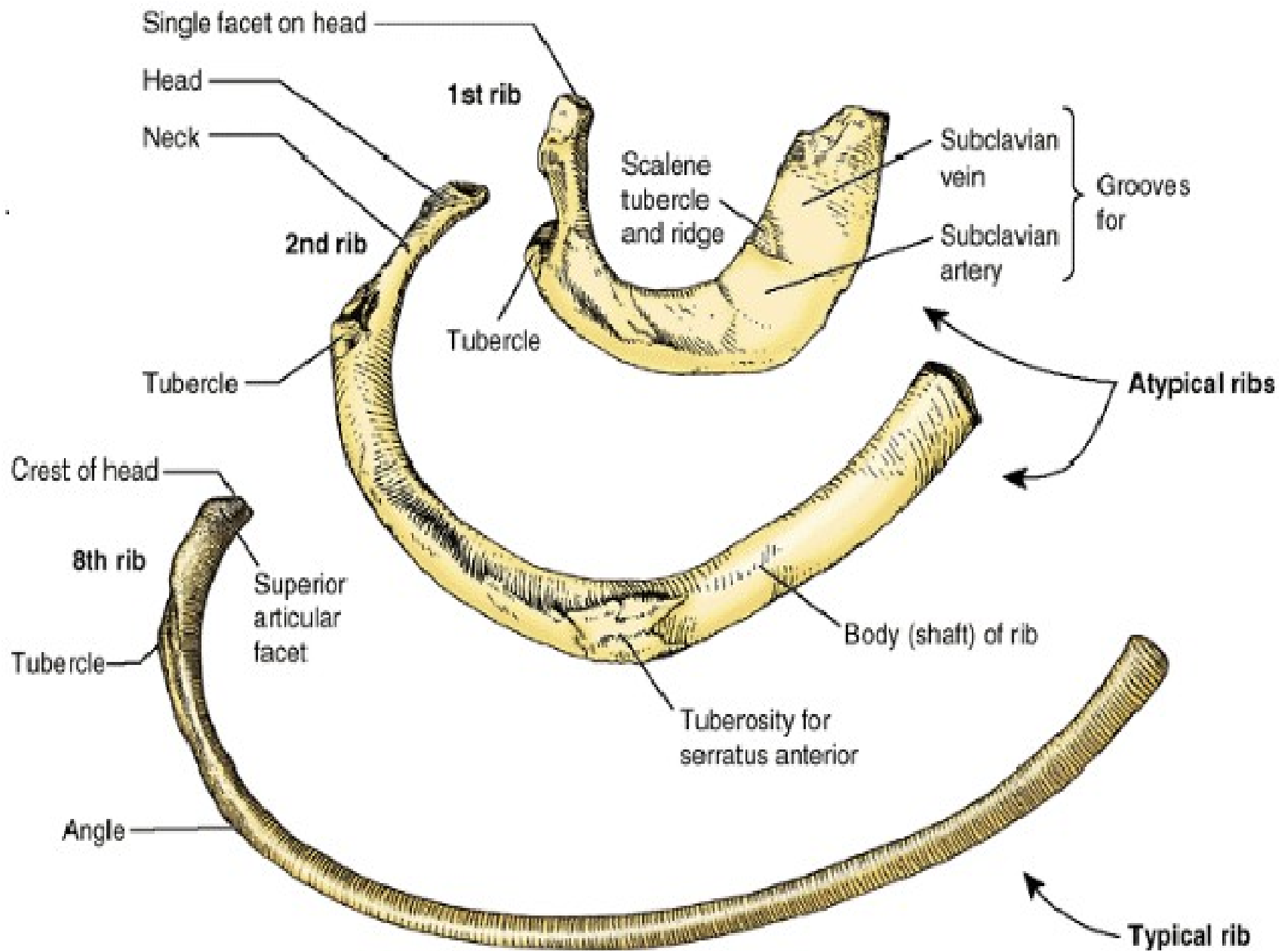
Atypical ribs (1st, 2nd, & 10th-12th) :

- **The 1st rib**: is the **broadest** (its body is widest & nearly horizontal), **shortest**, & most **sharply curved**
- it has two grooves crossing its superior surface for the **subclavian vessels**.
- The 1st rib has **single facet on its head**.
- **The 2nd rib**: is thinner, less curved & much longer than 1st rib.
- It has two facets on its head for articulation with the bodies of the T1 and T2 vertebrae.
- It has tuberosity for **serratus anterior muscle**
- **The 10th -12th ribs**, like the 1st rib, have **only one facet** on their heads.
- **The 11th and 12th ribs** are short and have **no necks or tubercles**.

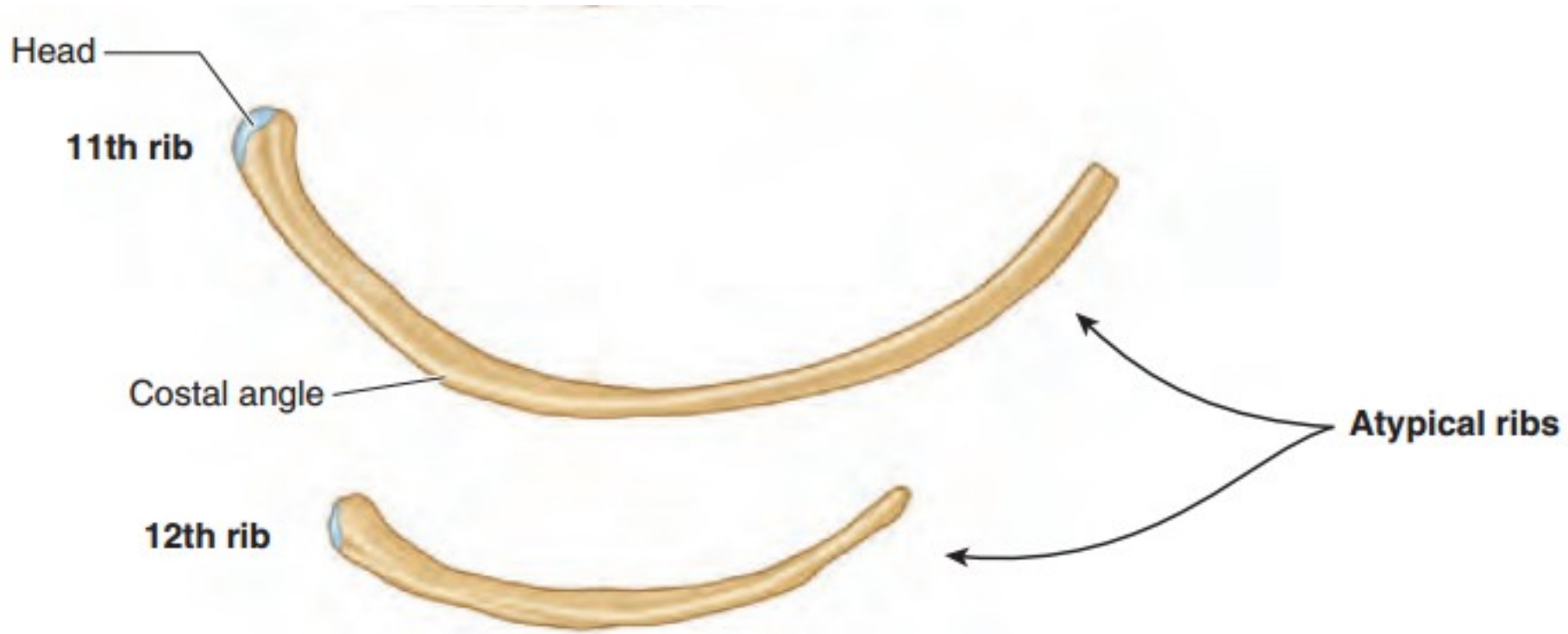
Atypical ribs

Rib I





Atypical Ribs

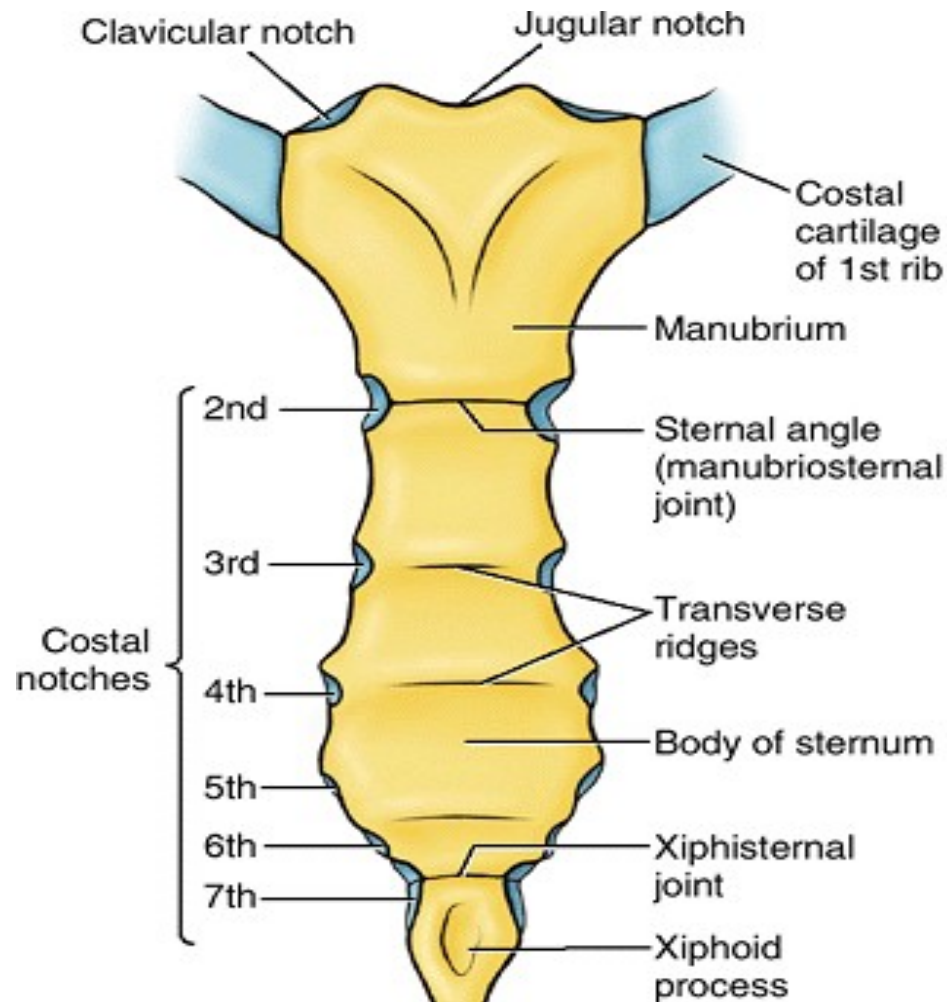


Intercostal spaces

- Separate the ribs and their costal cartilages from one another.
- Spaces and neurovascular structures are named according to the rib forming the **superior border of the space**. Eg. The 3th intercostal space lies between ribs 3 and 4.
- These spaces are occupied by intercostal muscles and intercostal blood vessels and nerves.
- There are **11 intercostal** spaces and **11 intercostal** nerves.
- The **subcostal space** is below the 12th rib and the anterior ramus of spinal nerve 12 is the **subcostal nerve**.

The Sternum(Breast bone)

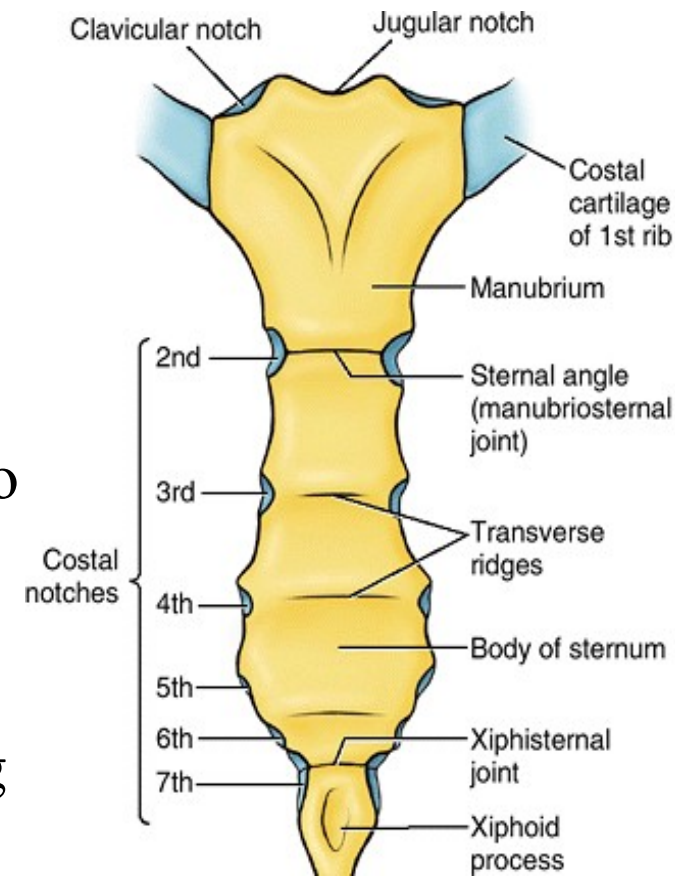
- is the flat, elongated bone that forms the middle of the anterior part of the thoracic cage.
- It consists of three fused bones
 - **Manubrium**
 - **Body**
 - **Xiphoid process**



The Sternum cont..

- **Manubrium**

- Located at level of **T3-T4**
- Wide superiorly and narrow inferiorly
- Center of the superior border of the manubrium is the **jugular notch** (suprasternal notch).
- **Clavicular notch** articulate with clavicle
- Inferolateral to the clavicular notch, the costal cartilage of the 1st rib is attached to the lateral border of the manubrium
- **Manubriosternal joint** is a junction between **manubrium** and **body** of sternum; their junction forms a projecting **sternal angle** (angle of Louis).



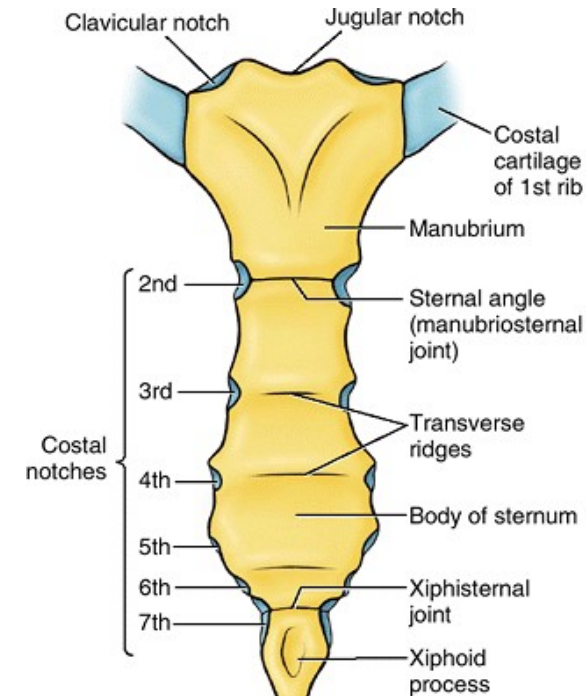
The Sternum cont..

- **Body**

- Located at level of **T5-T9**
- Lateral wall has **costal notches**
- Marked by **3 transverse ridges** – line of fusion of its four originally separate sternebrae.

- **Xiphoid process**

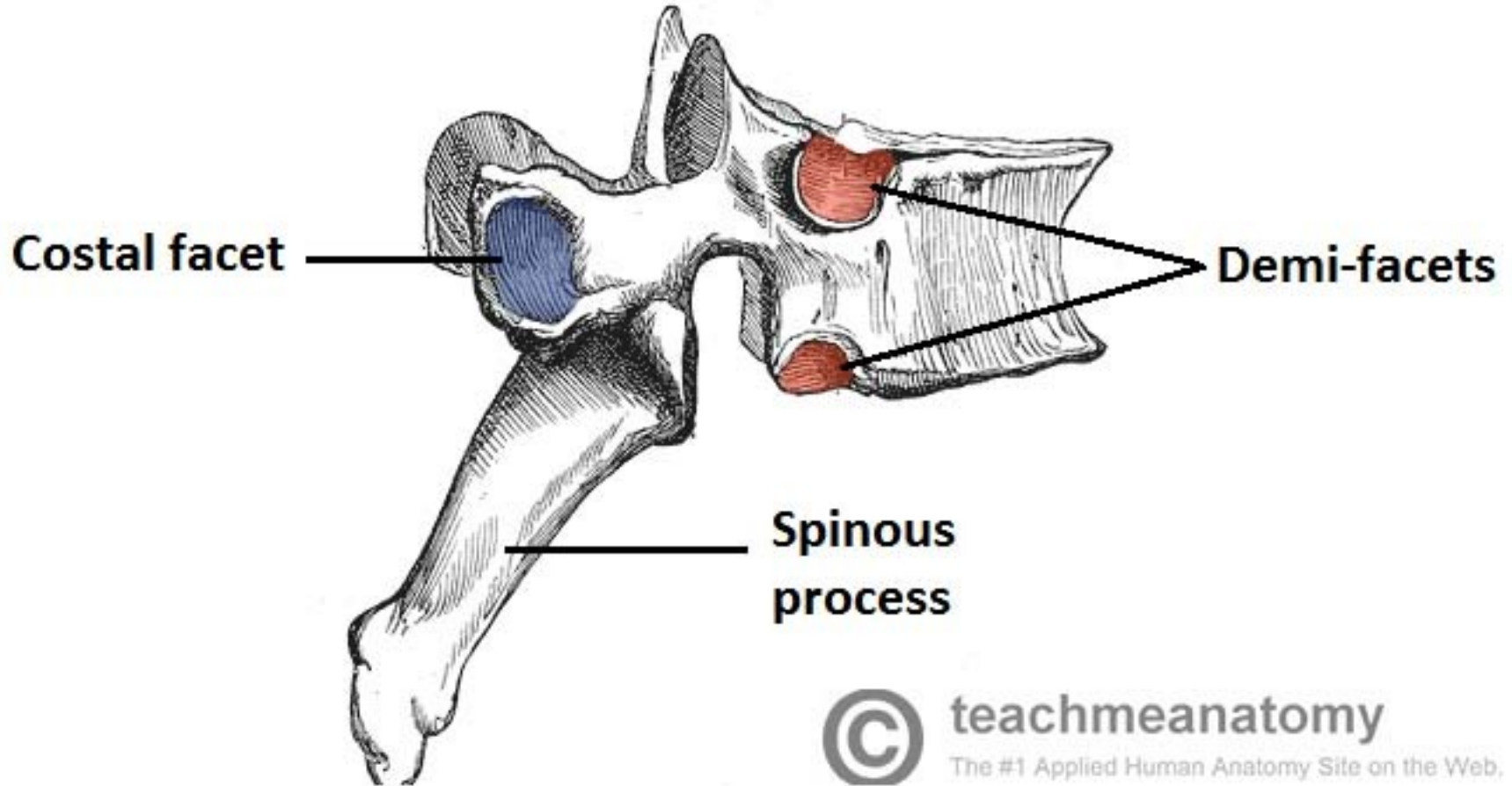
- the smallest and most variable part of the sternum, is thin and elongated.
- Its **inferior** end lies at the level of **T10** vertebra
- It is **cartilaginous in young people** but more or less ossified in adults older than age 40.
- **Landmark:** It is a midline marker for the **superior limit of the liver**, the central tendon of the **diaphragm**, and the inferior border of the **heart**



Thoracic vertebrae

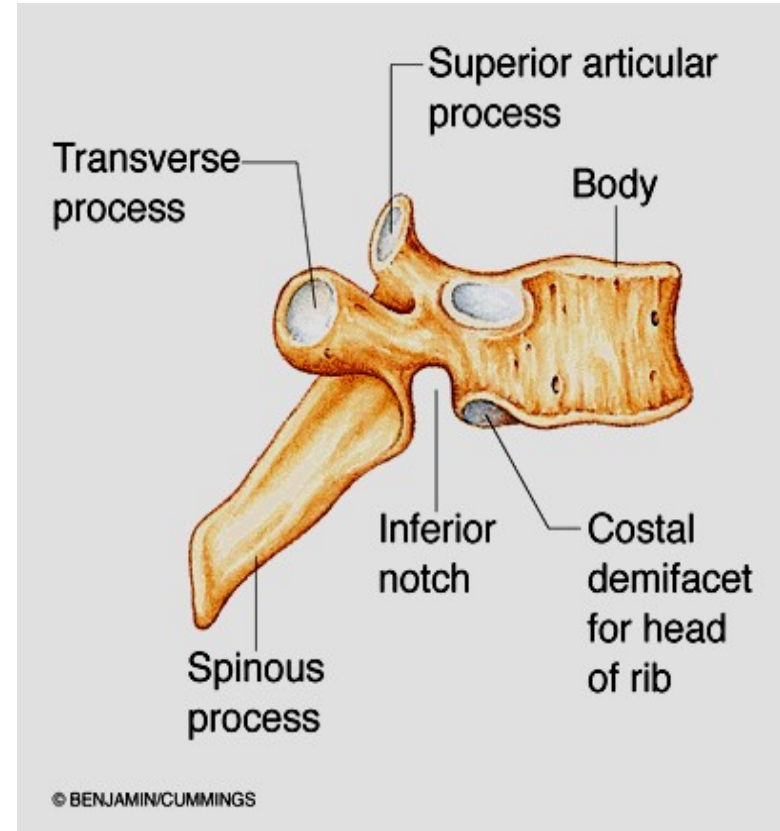
- Most **thoracic vertebrae** are typical in that they are independent, have **bodies**, vertebral **arches**, and seven **processes** for muscular and articular connections.
- *Characteristic features of thoracic vertebrae*
 - Bilateral *costal facets* (*demifacets*) on the vertebral bodies for articulation with the heads of **ribs**.
 - *Costal facets* on the *transverse processes* for articulation with the **tubercles** of ribs, except for the T10, T11 and T12.
 - The **spinous processes** are long and slant inferiorly.
 - This offers increased protection to the **spinal cord**, preventing an object such as a knife entering the spinal canal

Thoracic vertebrae con't...

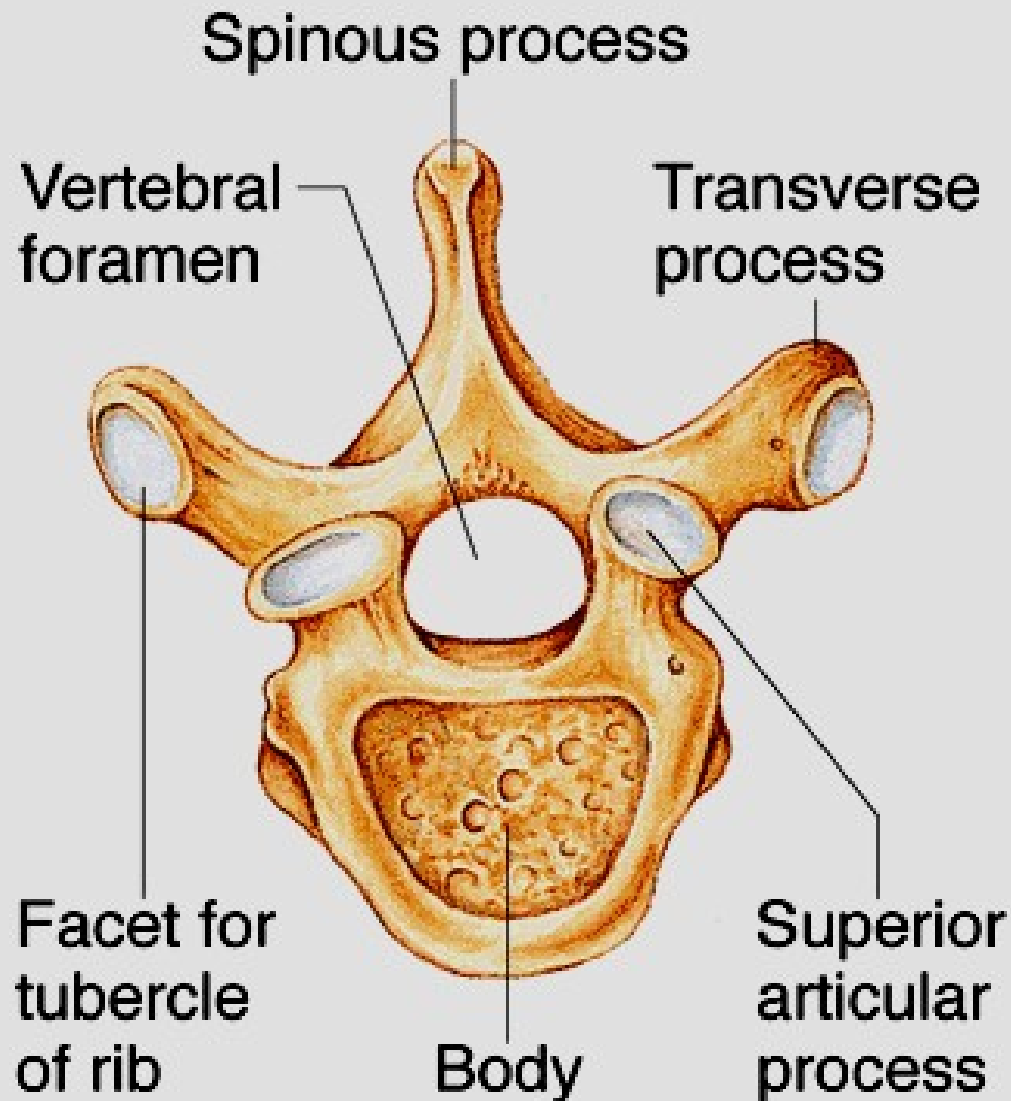


Thoracic Vertebrae con't...

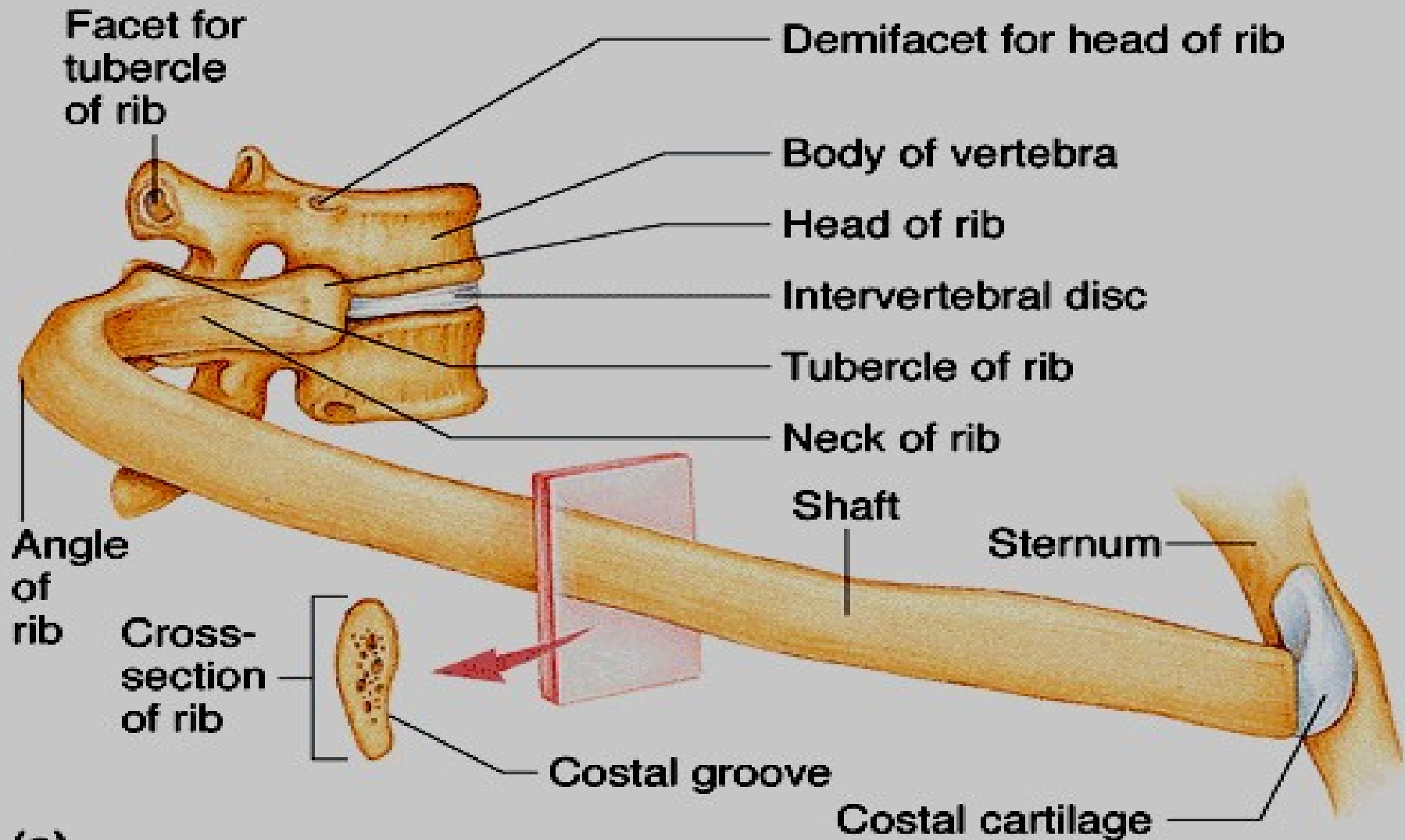
- **Body** is larger than cervical; heart shaped
- **Spinous process** is long and sharp, projects inferiorly
- **Vertebral foramen** is circular
- **Transverse processes** project posteriorly and bear costal facets for ribs (T1-T10)
- **Body bears two costal demifacets (T2-T9)**
 - **superior** – articulate with head of its own rib
 - **Inferior**- articulate with head of the rib inferior to it
- **T1,T10-T12:** one facet on the body



Thoracic Vertebrae con't...



Thoracic Vertebrae

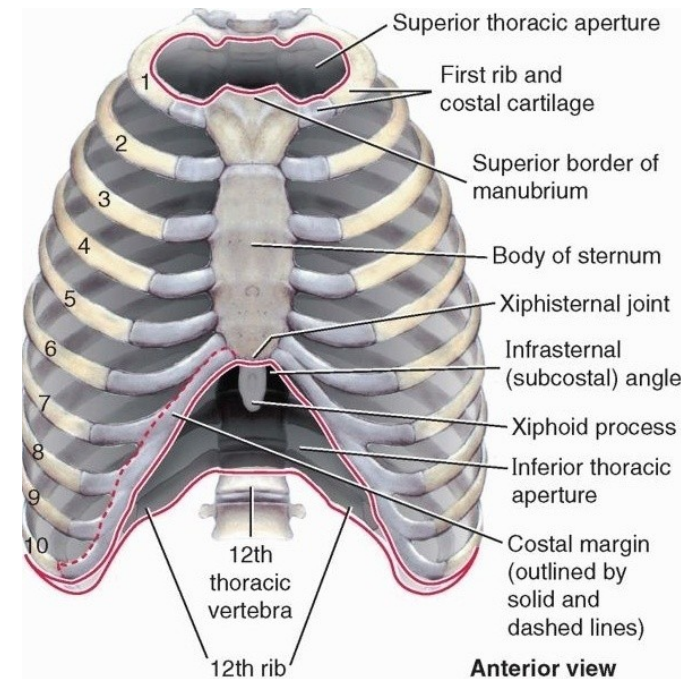


(a)

Thoracic apertures

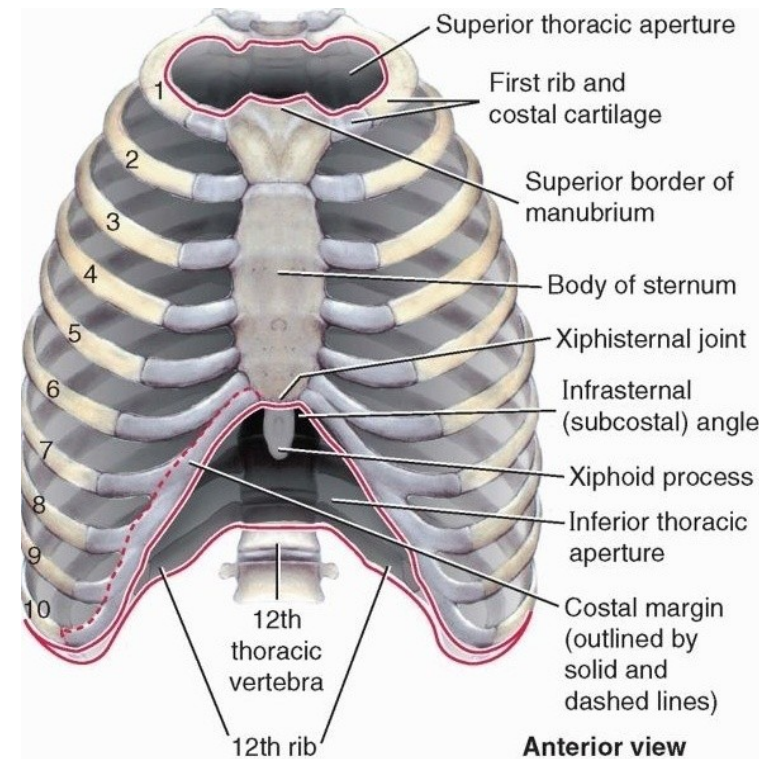
Superior thoracic aperture (*thoracic inlet*)

- It is the site of passage of structures between the thoracic cavity and the neck through it
- is bounded
 - **Posteriorly**- by vertebra T1
 - **Laterally**- by the **1st pair** of ribs and their **costal cartilages**.
 - **Anteriorly**- by the superior border of the **manubrium**.



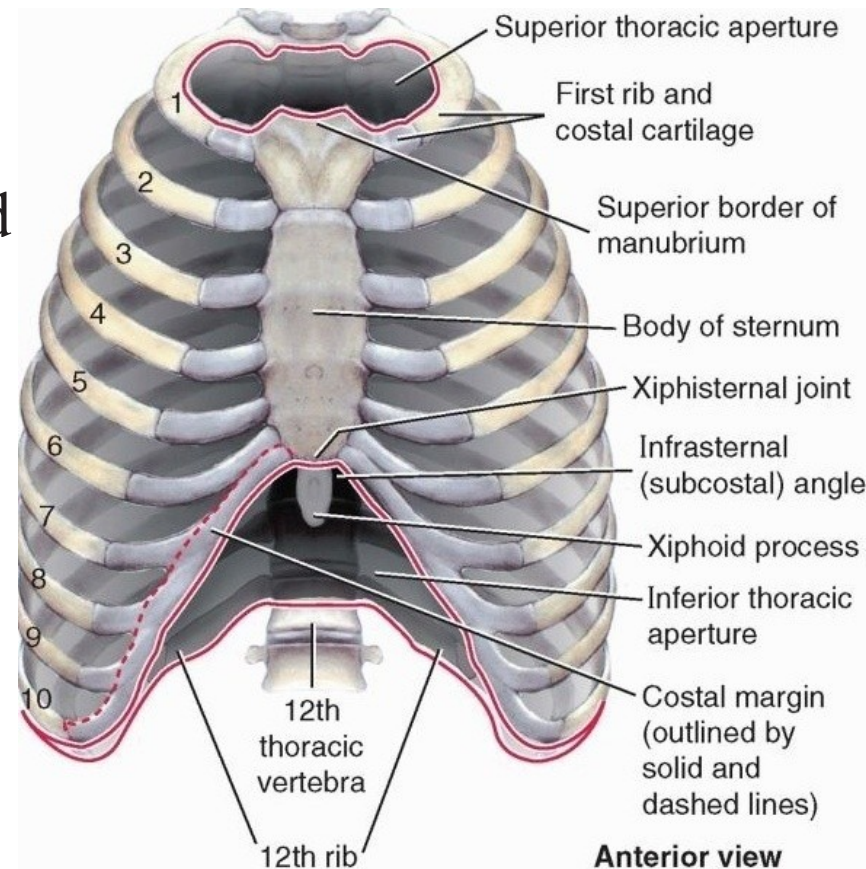
Thoracic apertures

- **Inferior thoracic aperture (*thoracic outlet*)**
- is irregular in outline
- By closing the inferior thoracic aperture, diaphragm separates the **thoracic** and **abdominal** cavities
- Structures passing from or to the thorax or from the abdomen pass through openings that traverse the diaphragm eg. diaphragm pierced by the inferior vena cava at (T8), aorta at (T12) and esophagus at (T10)



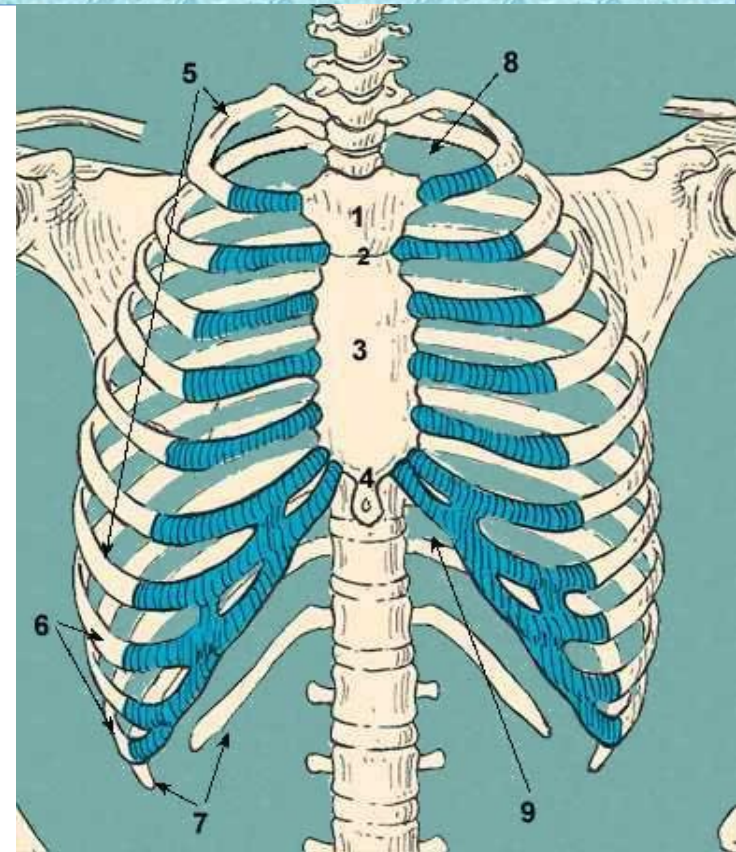
Inferior thoracic aperture...

- Is **bounded** by :
 - **Posteriorly**- by the **T12**.
 - **Posterolaterally**- by the **11th** and **12th** pairs of ribs.
 - **Anterolaterally**- by the joined **costal cartilages** of ribs **7–10** forming the costal margins.
 - **Anteriorly**- by the xiphisternal joint.



Joints of thoracic wall

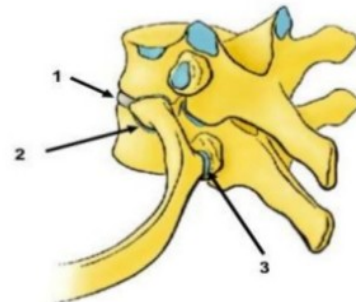
- ❑ **Intervertebral joint** : *joints* b/n the bodies of adjacent vertebrae
- ❑ **Costovertebral joint**: joints b/n ribs and vertebrae
- ❑ **Costotransverse joints**: joints B/n transverse process of vertebra with tubercle of ribs.
- ❖ Sternocostal joint
- ❖ Sternoclavicular joint
- ❖ Costochondral joint
- ❖ Interchondral joint
- ❖ Manubrosternal joint
- ❖ Xiphisternal joint



Joints of Thoracic Cage

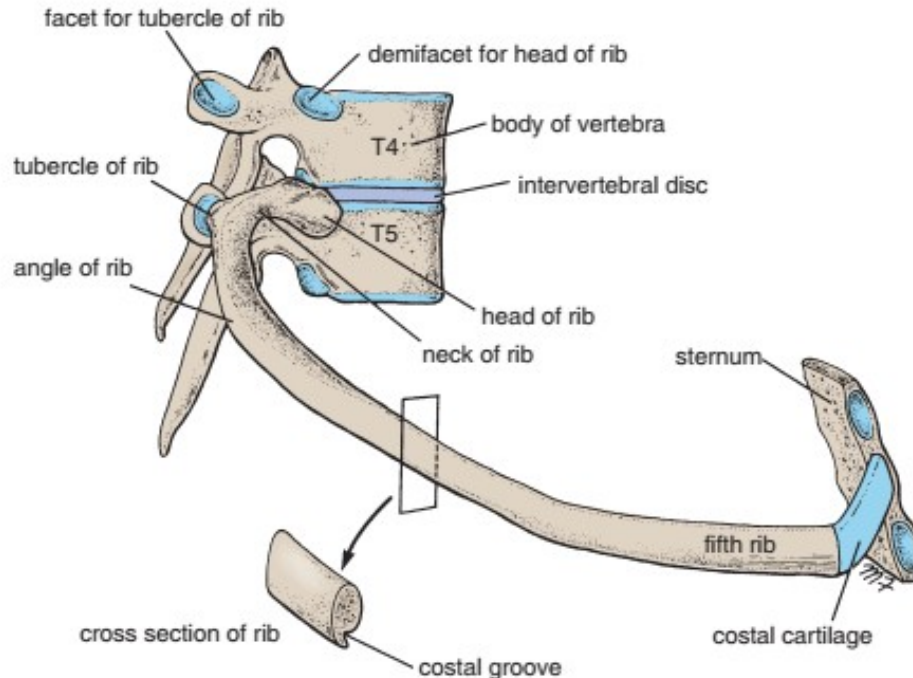
Posteriorly:

1. Intervertebral joints
(2° cartilaginous)
2. Costovertebral joints
(synovial plane)
3. Costotransverse joints
(synovial plane)



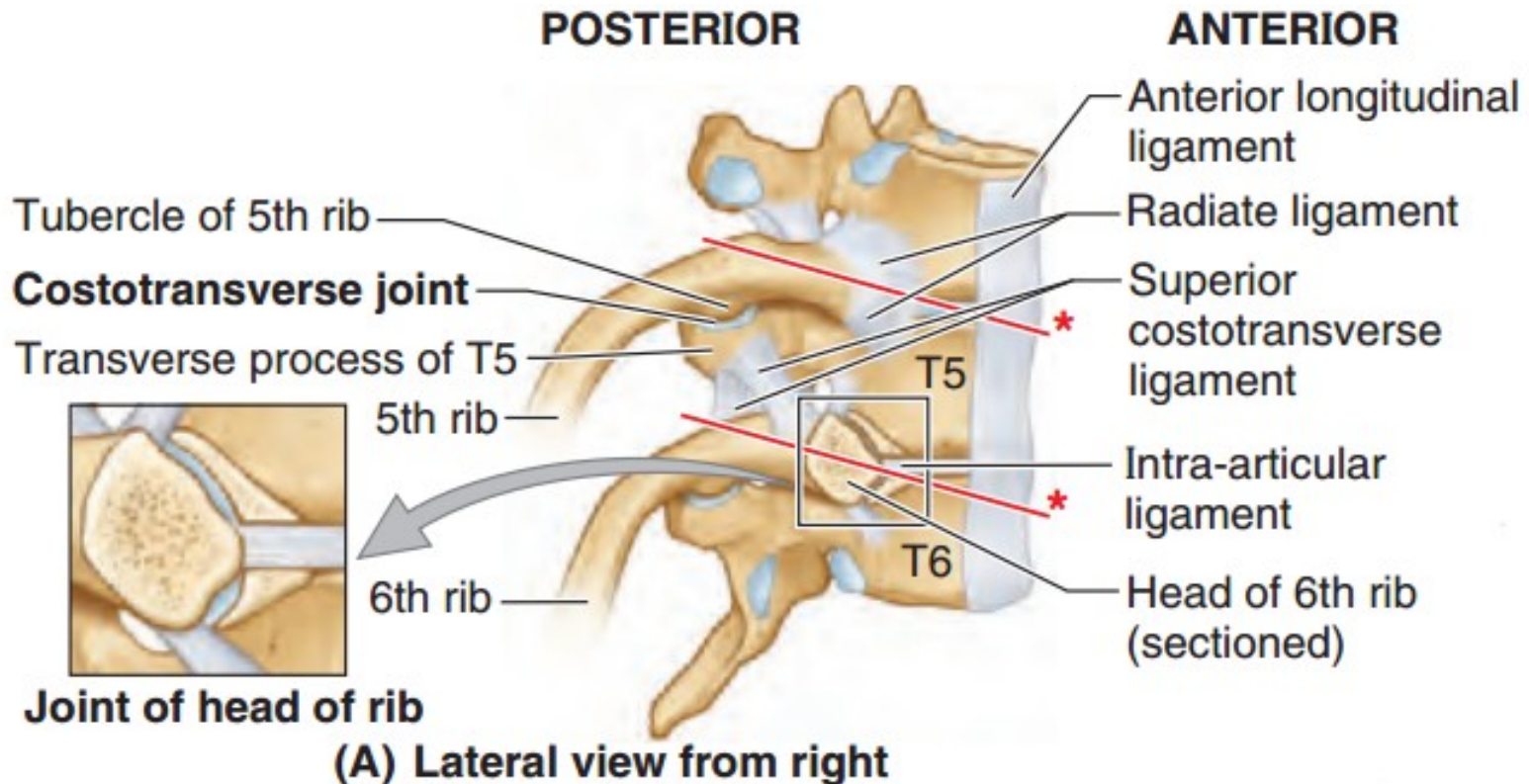
Costovertebral joint

- A typical rib articulates posteriorly with the vertebral column at two joints,
 - the joints of heads of ribs and
 - costotransverse joints



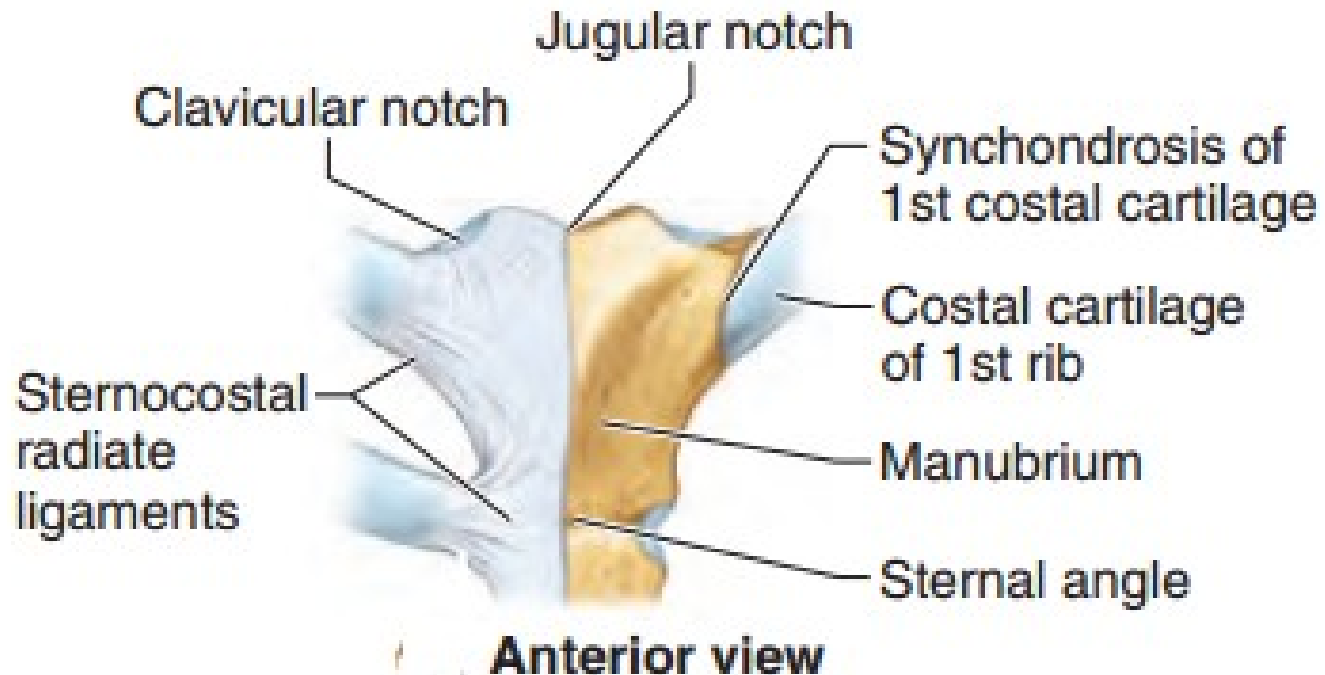
Costovertebral Joints

- Joints B/n transverse process of vertebra with tubercle of ribs.
- Abundant ligaments lateral to the posterior parts of the vertebrae provide strength to and limit the movements of these joints.



Sternocostal joint

- The 1st pair of costal cartilages articulates with the manubrium by means of thin dense fibrocartilage.
- The 2nd–7th pairs of costal cartilages articulate with the sternum at **synovial joints** allowing **movement during respiration**.



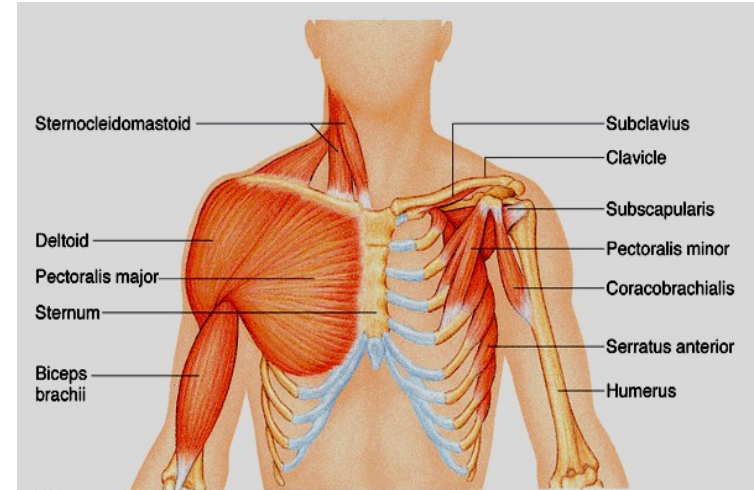
Con't...

- **Interchondral** : Between **costal cartilages** of 6th and 7th, 7th and 8th, and 8th and 9th ribs
- **Sternoclavicular**: Sternal end of **clavicle** with **manubrium** of sternum and 1st costal cartilage
- **Manubriosternal**: Articulation between **manubrium** and **body** of sternum
- **Xiphisternal**: Articulation between **xiphoid** process and **body** of sternum

Muscles of the thoracic wall

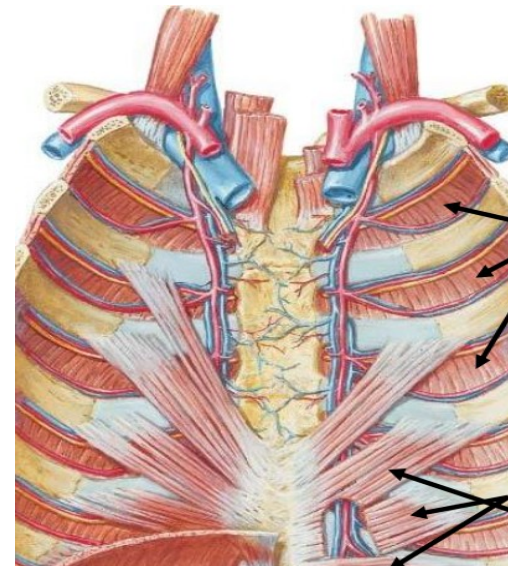
➤ Muscles related to thoracic wall

- Pectoral muscles
- Muscles of abdomen
- Latissimus dorsi
- Serratus anterior
- Rhomboid muscles



➤ The true muscles of the thoracic wall

- Serratus posterior
- Levator costarum
- Intercostals thoracic
- Subcostals thoracic
- Transversus thoracic



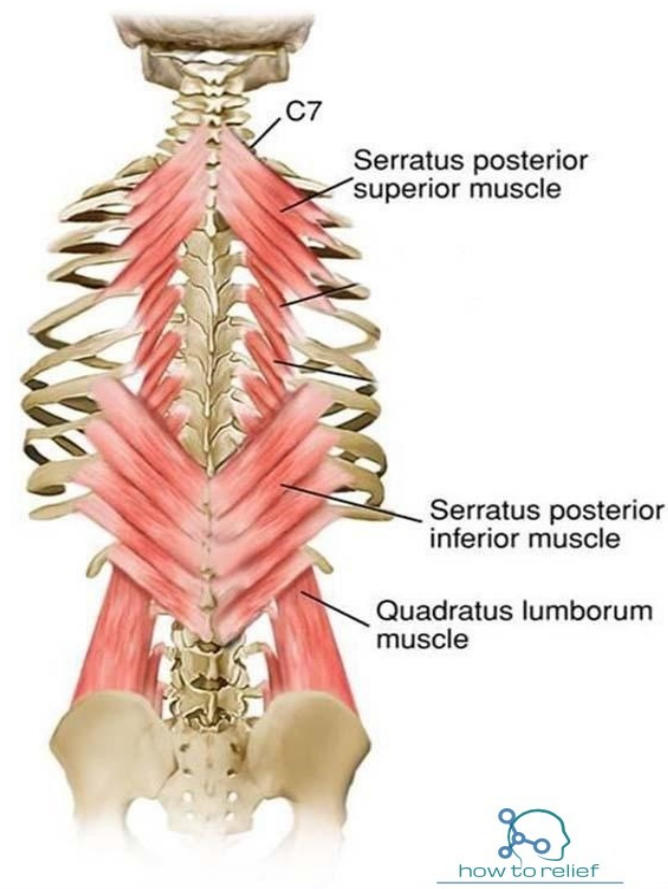
Muscles of thorax proper: **Serratus posterior**

- **Serratus posterior superior**

- ✓ **Origin** - ligamentum nuchae and spines of C7-T3
- ✓ **Insertion** - superior border of 2-4 ribs
- ✓ **Action** - **elevates** superior ribs
- ✓ **Inervation** - 2nd to 5th intercostal nerves

- **Serratus posterior inferior**

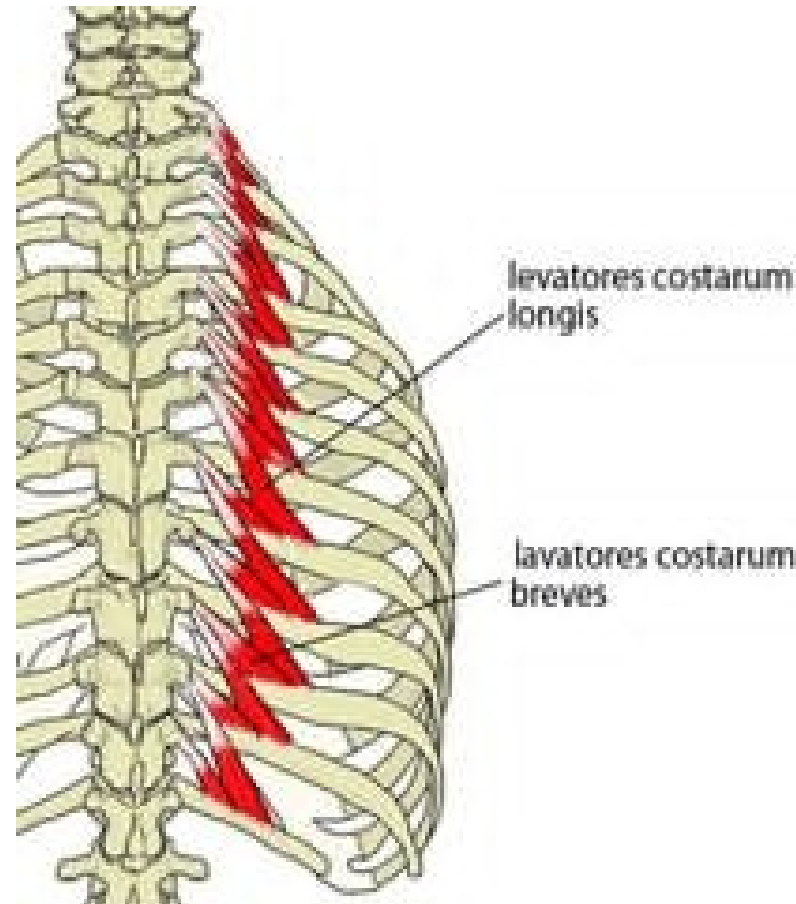
- ✓ **Origin** - spines of T11-L2
- ✓ **Insertion** - inferior border of inferior 8th to 12th ribs
- ✓ **Action** - **depress** inferior ribs
- ✓ **Innervation** : **Anterior rami T9 to T12 thoracic nerves**



Muscles of thorax proper con't...

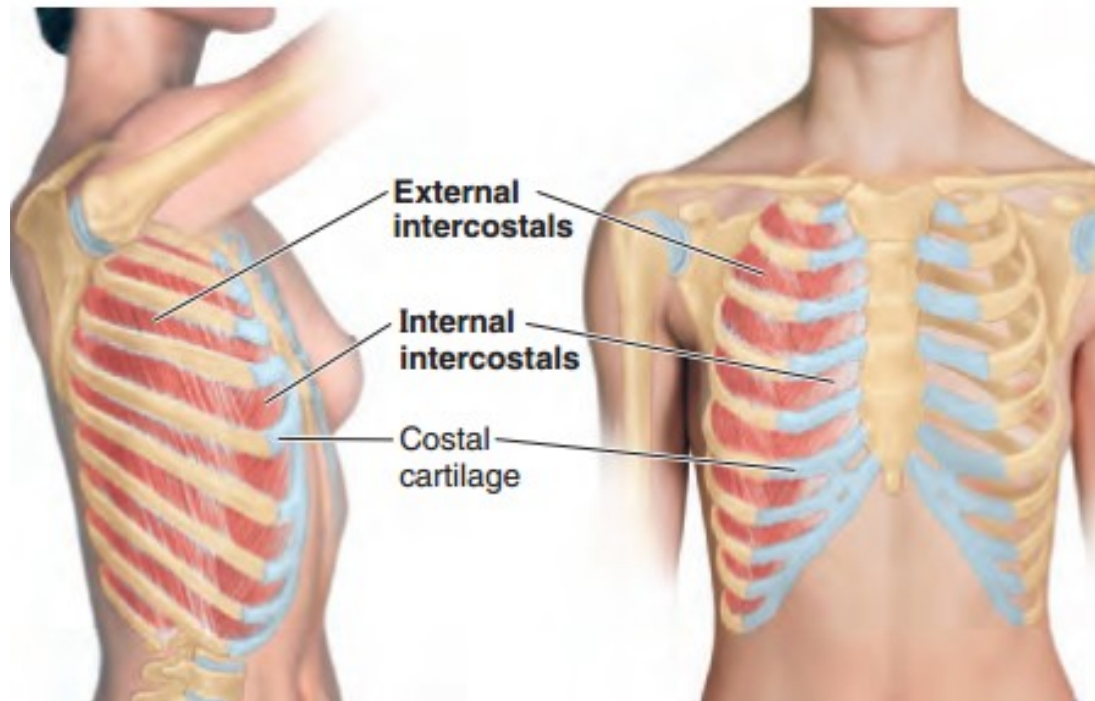
- **Levator costarum**

- **Origin** – transverse processes of C7 –T11
- **Insertion** - ribs between tubercle and angle
- **Inervation** - **Posterior primary rami of C8–T11 nerves**
- **Action** - elevates ribs



Muscles of thorax proper con't...

- **The three layers of the intercostal muscles are:**
 - **External layer** -- external intercostal
 - **Internal layer** -- internal intercostal
 - **Innermost layer** -- transversus thoracic (anterior), innermost (lateral) and subcostal (posterior).

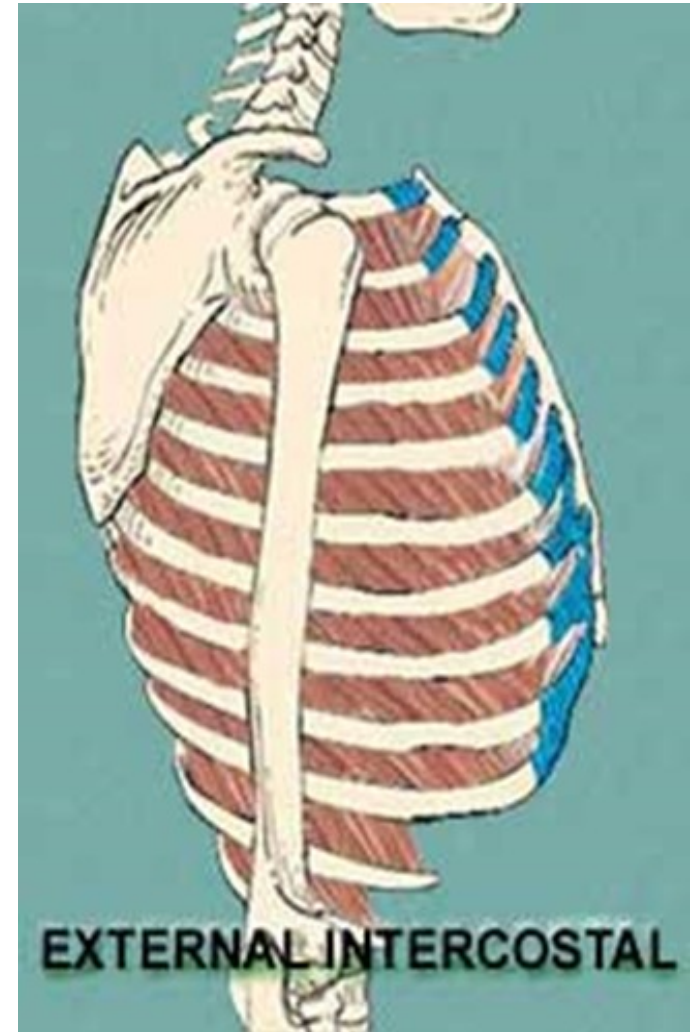


(B) Lateral view

(C) Anterior view

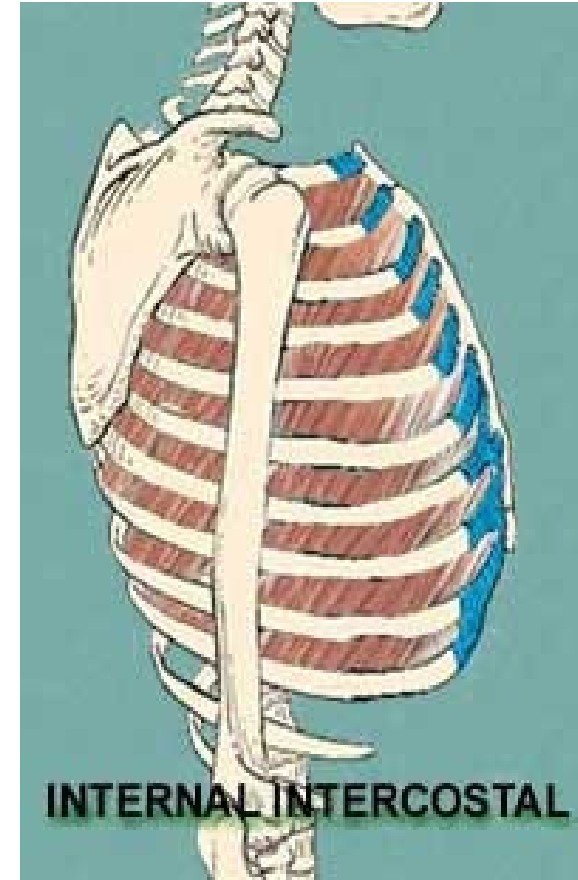
External intercostal muscles

- They occupy the intercostal spaces from the **tubercles** of the ribs posteriorly to the **costochondral** junctions anteriorly.
- **Anteriorly**, the muscle fibers are replaced by the **external intercostal membranes**.
- These muscles run **infero-anteriorly** from the rib above to the rib below.
- These muscles are continuous inferiorly with the *external oblique muscles* in the anterolateral abdominal wall.
- **Inervation** - **Intercostal nerve**
- **Action** - Elevate ribs during forced inspiration



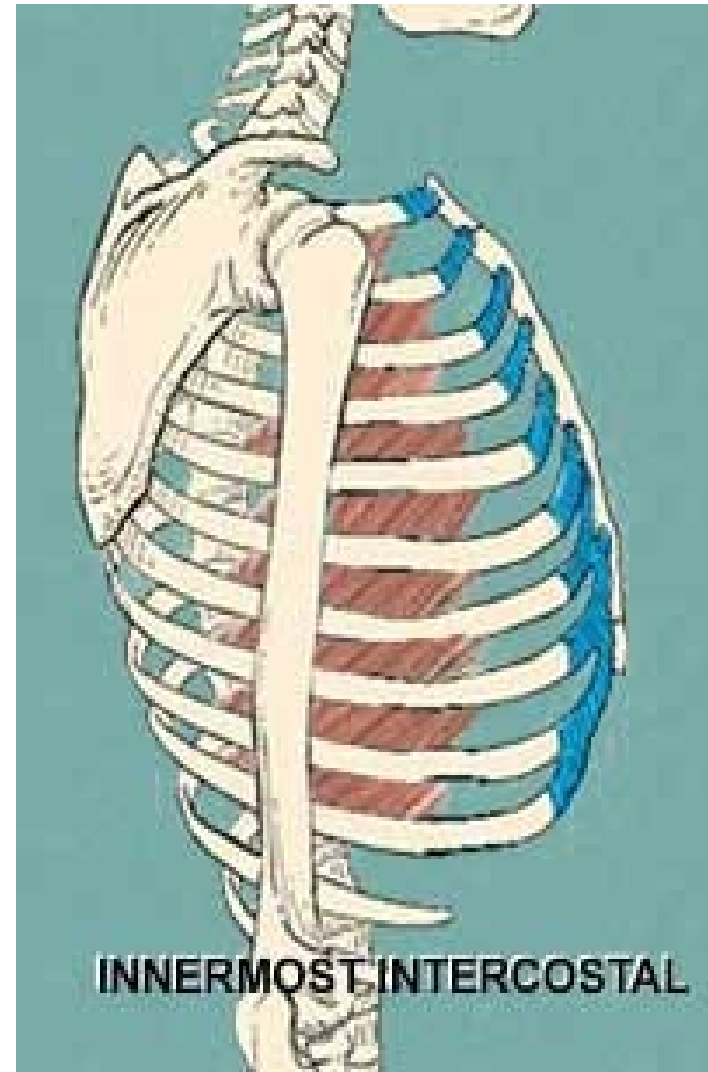
Internal intercostal muscles

- Run deep to and at right angles to the external intercostal muscles.
- Their fibers run **infero-posteriorly** from the floors of the costal grooves to the superior borders of the ribs inferior to them.
- Between the ribs **posteriorly**, medial to the angles, the internal intercostals are replaced by the **internal intercostal membranes**.
- The inferior internal intercostal muscles are continuous with the **internal oblique muscles** in the anterolateral abdominal wall.
- **Inervation** - **Intercostal nerve**
- **Action**- **Interosseous part**: depresses ribs
 - **Interchondral part**: elevates ribs during forced respiration



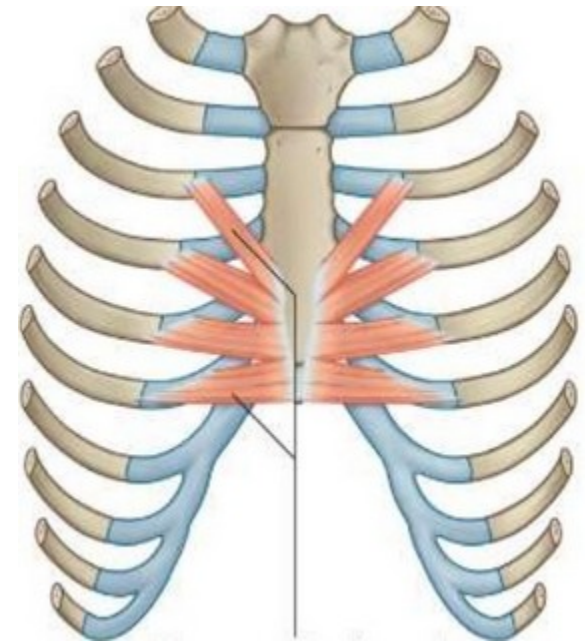
Innermost intercostal muscles

- are similar to the internal intercostals.
- These muscles pass between the internal surfaces of adjacent ribs and occupy the **lateral-most parts** of the intercostal spaces.
- The innermost intercostals are separated from the internal intercostals by **intercostal nerves and vessels**.
- It is likely that their actions are the same as those of the internal intercostal muscles.



Transversus thoracis muscles

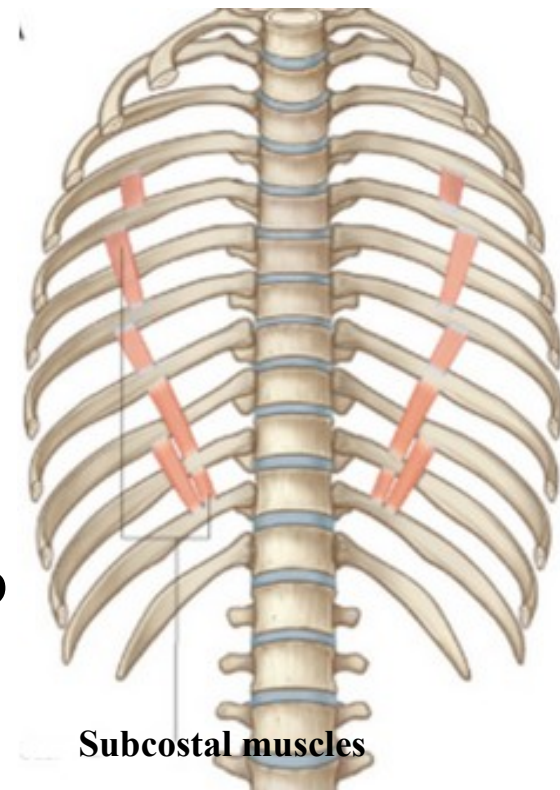
- Consist of four or five slips that radiate superolaterally from the posterior aspect of the inferior sternum.
- The transversus thoracis muscles are continuous inferiorly with the *transversus abdominis muscles* in the anterolateral body wall.
- These muscles appear to have a **weak expiratory** function
- **Origin** - Posterior surface of lower sternum
- **Insertion** -Internal surface of costal cartilages 2–6
- **Inervation**- **Intercostal nerve**
- **Action**- Weakly depress ribs



Transversus thoracis

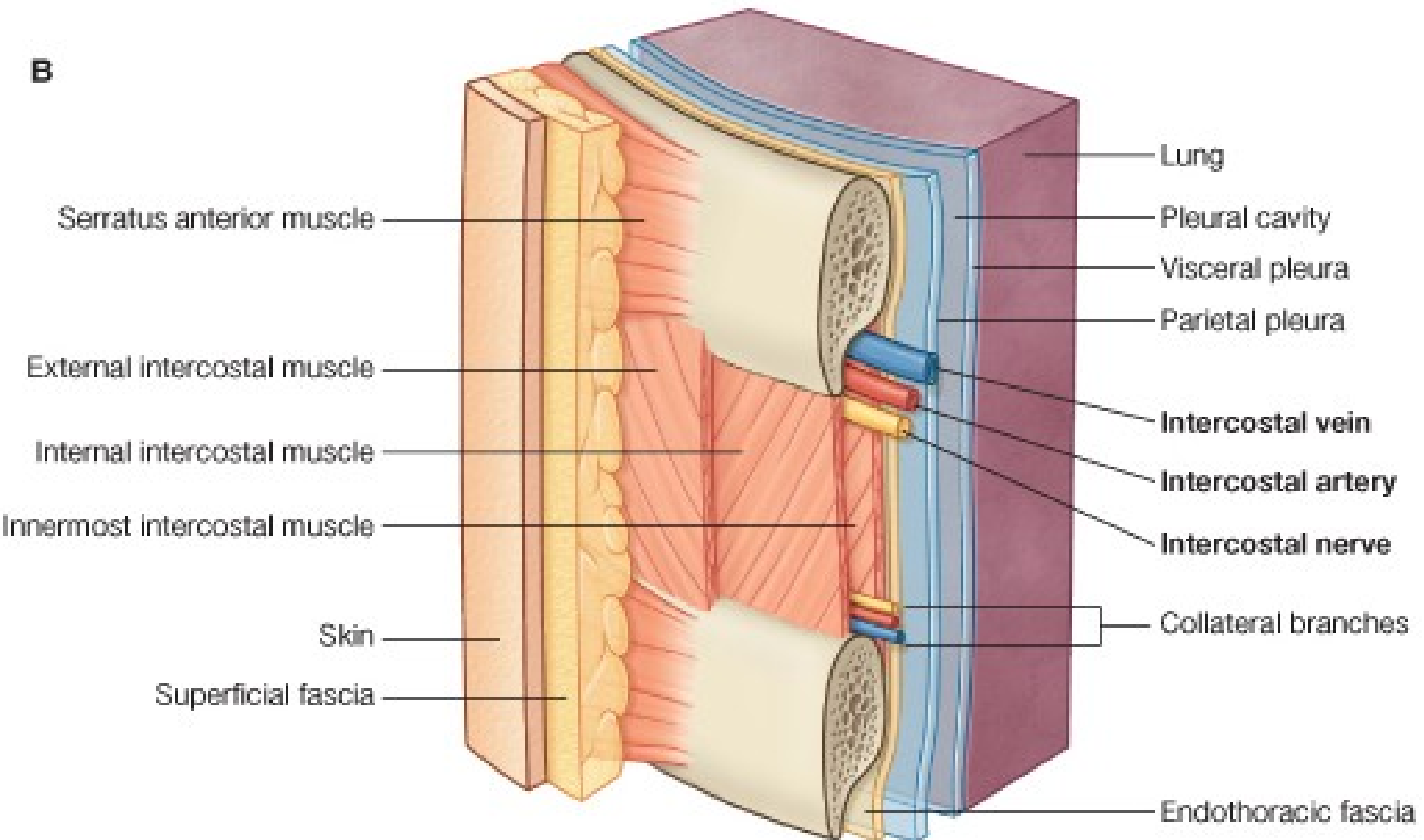
Subcostal muscles

- Are variable in size and shape, usually well developed only in the lower thoracic wall.
- These thin muscular slips extend from the **internal surface** of the **angle** of one rib to the internal surface of the second or third rib inferior to it.
- Crossing one or two intercostal spaces, the subcostals run in the **same direction as the innermost intercostals** and blend with them.



Muscles of the Thoracic Wall

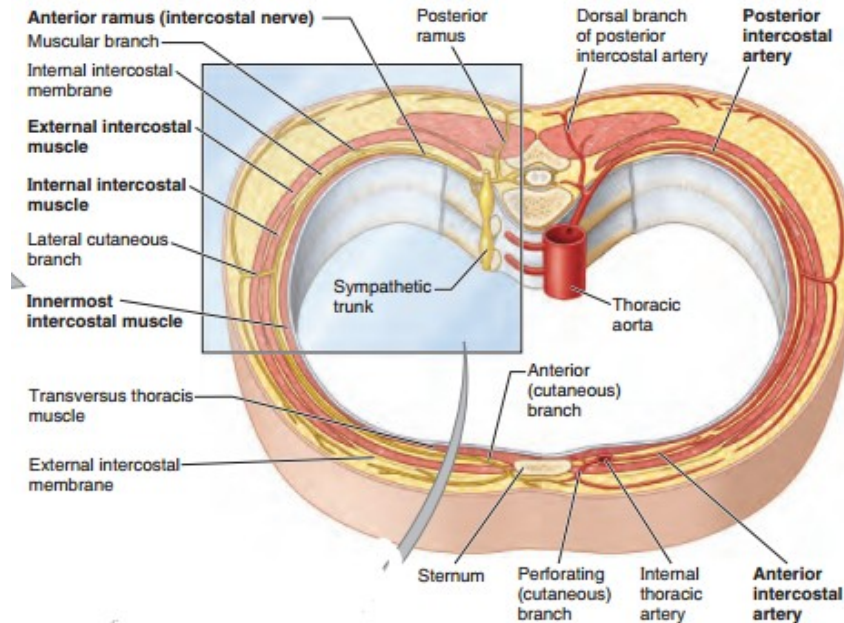
B



Muscle	Superior attachment	Inferior attachment	Innervation	Function
External intercostal	Inferior margin of rib above	Superior surface of rib below	Intercostal nerves; T1-T11	Most active during inspiration; supports intercostal space; moves ribs superiorly
Internalintercostal	Lateral edge of costal groove of rib above	Superior surface of rib below deep to the attachment of the related external intercostals	Intercostal nerves; T1-T11	Most active during expiration; supports intercostal space; moves ribs inferiorly
Innermost intercostal	Medial edge of costal groove of rib above	Internal aspect of superior surface of rib below	Intercostal nerves; T1-T11	Acts with internal intercostals muscles
Subcostales	Internal surface (near angle) of lower ribs	Internal surface of second or third rib below	Related inter-costal nerves	May depress ribs
Transversus thoracis	Lower margins and internal surfaces of cocostal cartilages of second to sixth ribs	Inferior aspect of deep surface of body of sternum, xiphoid process and costal cartilages ribs IV-VII	Related inter-costal nerves	Depresses costal cartilages

Nerves of thoracic wall

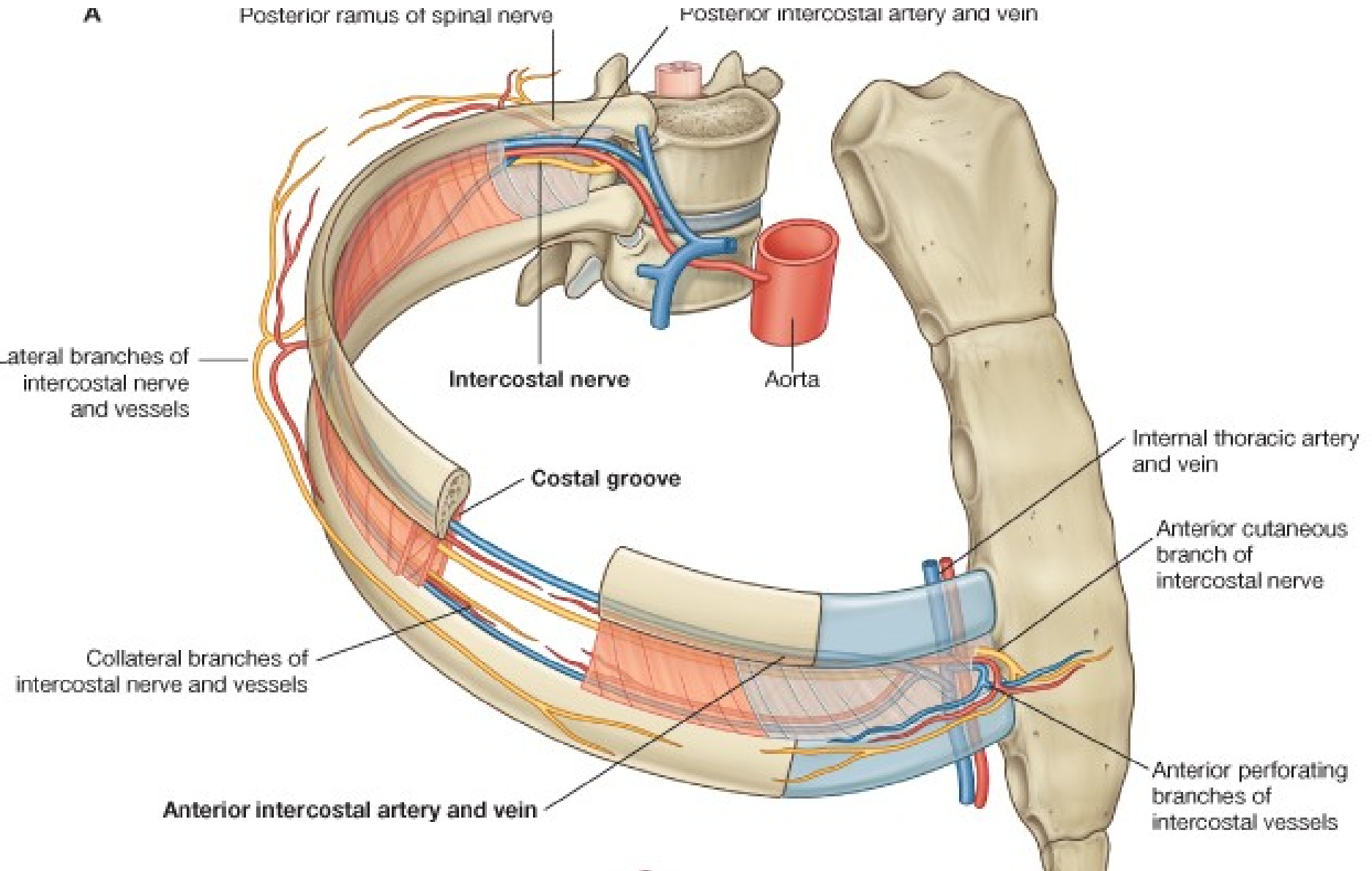
- **12 pairs** of thoracic spinal nerves
- Leave spinal cord through corresponding **intervertebral foramina** and divide into 2 branches
 - ✓ **Posterior (dorsal) rami:**
innervate muscles, bones, joints and skin of the **back**
 - ✓ **Anterior (ventral) rami:**
innervate **intercostal muscles**, periosteum of the ribs and skin of the thorax and abdomen



Course of typical intercostal nerves

- **Intercostal nerves** enter to posterior intercostal spaces, running between the **parietal pleura** and the **internal intercostal membrane**
- Near the **angles** of the ribs, the nerves run under **costal grooves** pass b/n the **internal IC** and the **innermost IC** muscles.
- The neurovascular bundles are thus **sheltered** by the inferior margins of the overlying ribs.
- **Collateral** branches of these nerves arise near the **angles** of the ribs and run along the superior border of the rib below.
- At **mid-axillary line IC nerve** giving rise to **lateral cutaneous branches**.
- Near the sternum, the intercostal nerves **turn anteriorly**, passing between the costal cartilages to become **anterior cutaneous branches**.

Course of typical intercostal nerves con't...



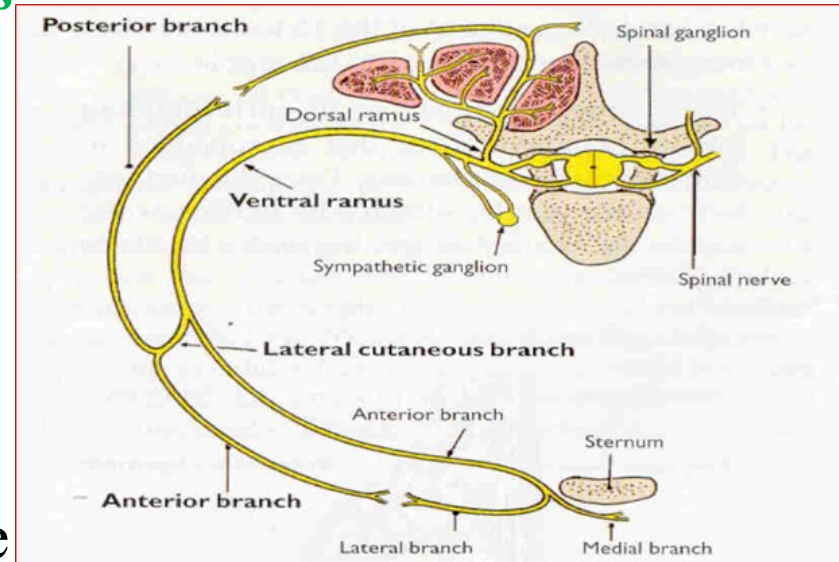
Branches of a typical intercostal nerve

- **Collateral branches**

- that arise near the angles of the ribs and
- descend to course along the **superior** margin of the **lower rib**,
- helping supply **intercostal muscles** and **parietal pleura**.

- **Lateral cutaneous branches**

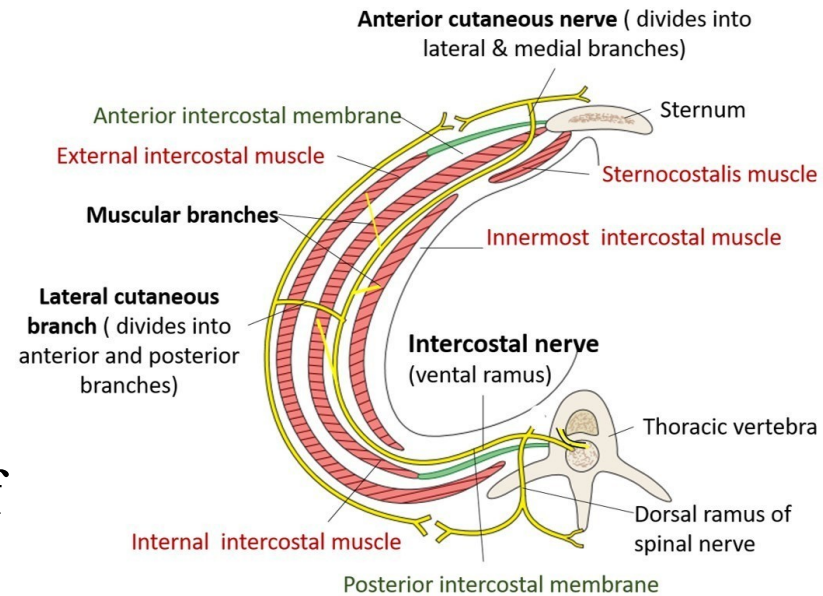
- that arise near the MAL
- pierce the internal and external intercostal muscles, and divide in turn into **anterior** and **posterior** branches.
- These terminal branches supply the **skin of the lateral thoracic and abdominal walls**.



Branches of a typical intercostal nerve...

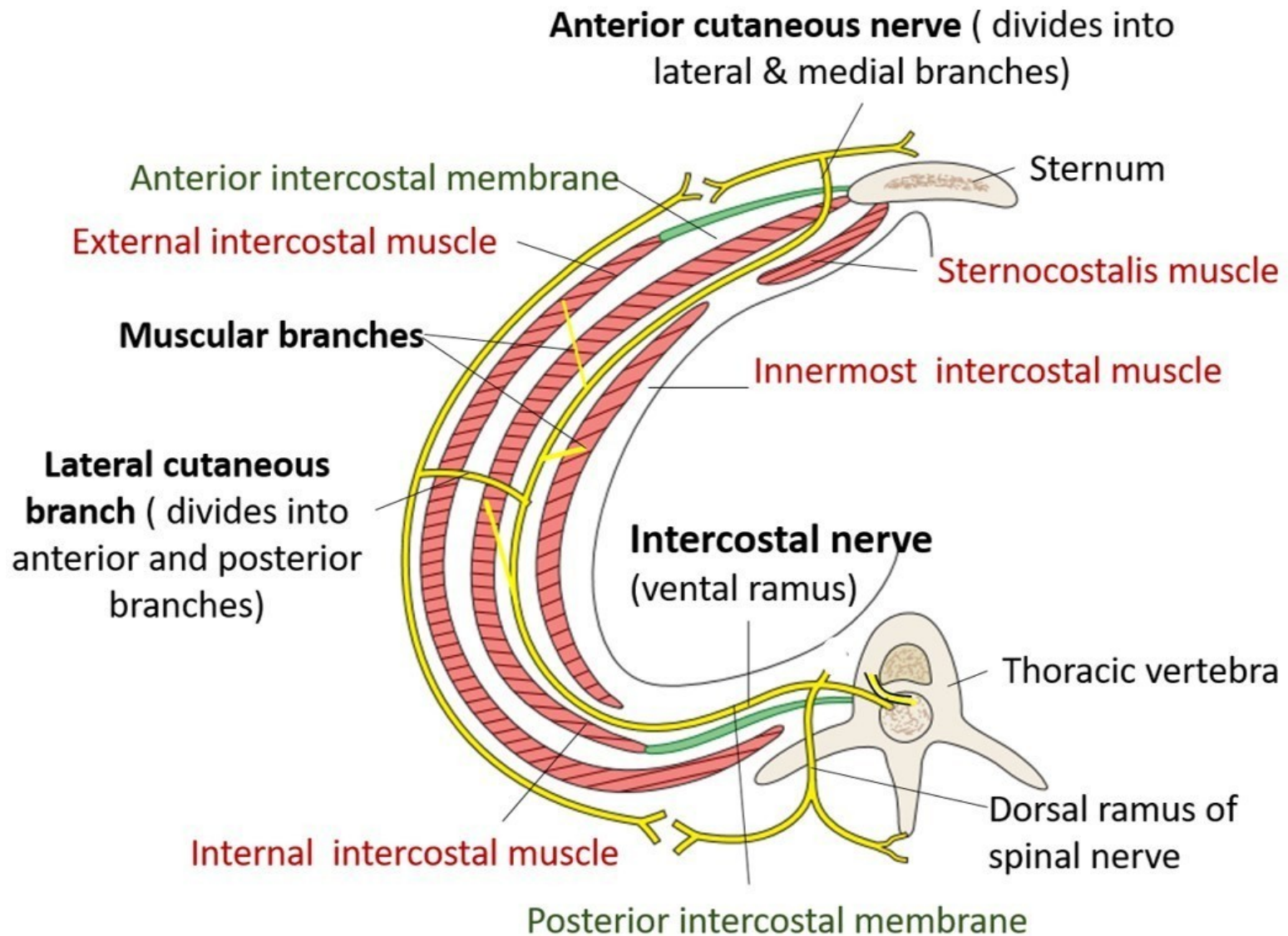
- **Anterior cutaneous branches**

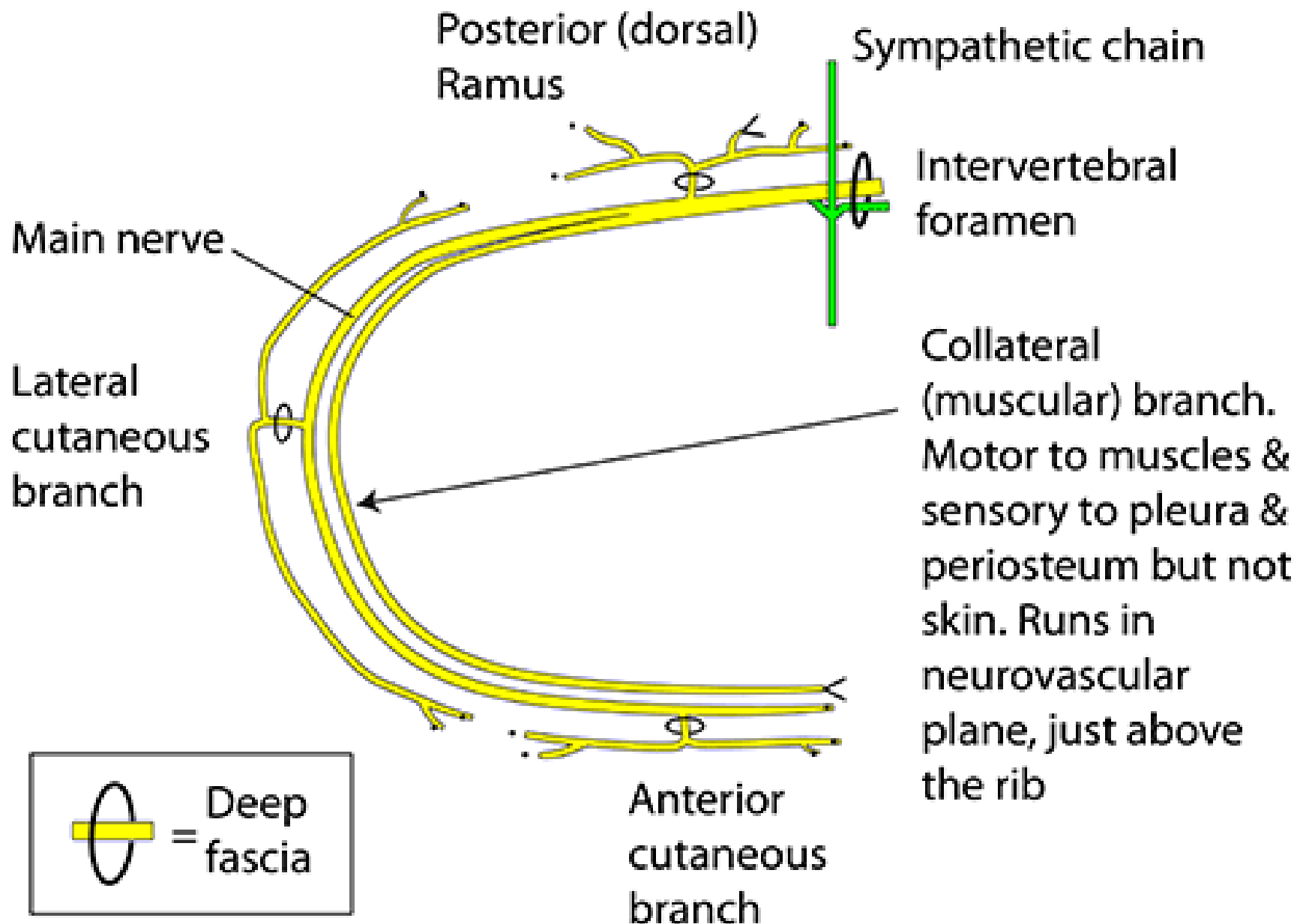
- pierce the muscles and membranes of the intercostal space in the parasternal line &
- divide into **medial** and **lateral** branches.
- These terminal branches supply the **skin** on the **anterior** aspect of the thorax and abdomen.



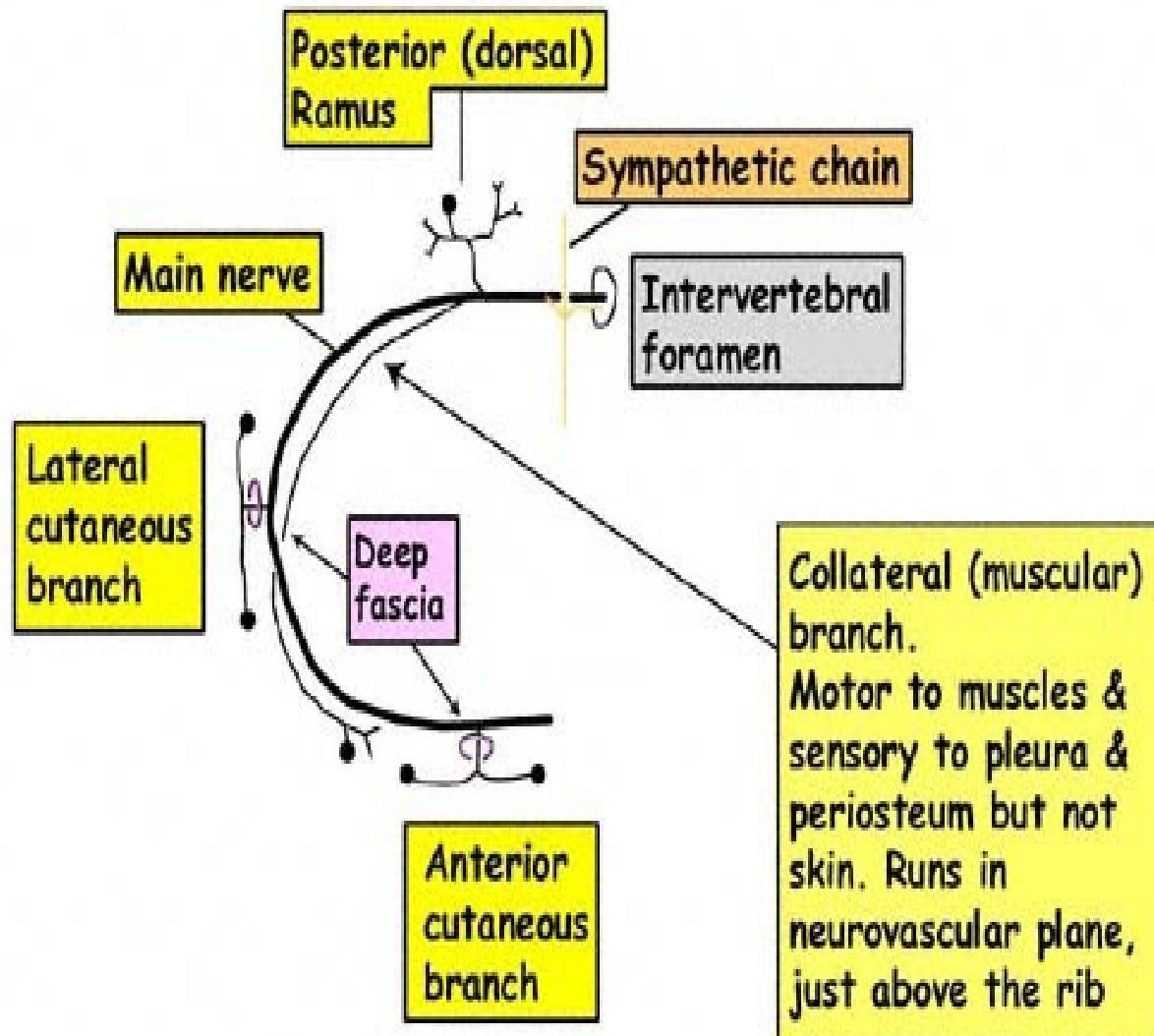
- **Muscular branches**

- that supply the intercostal, subcostal, transversus thoracis, levatores costarum, and serratus posterior muscles.





TYPICAL INTERCOSTAL NERVE



ATYPICAL INTERCOSTAL NERVES

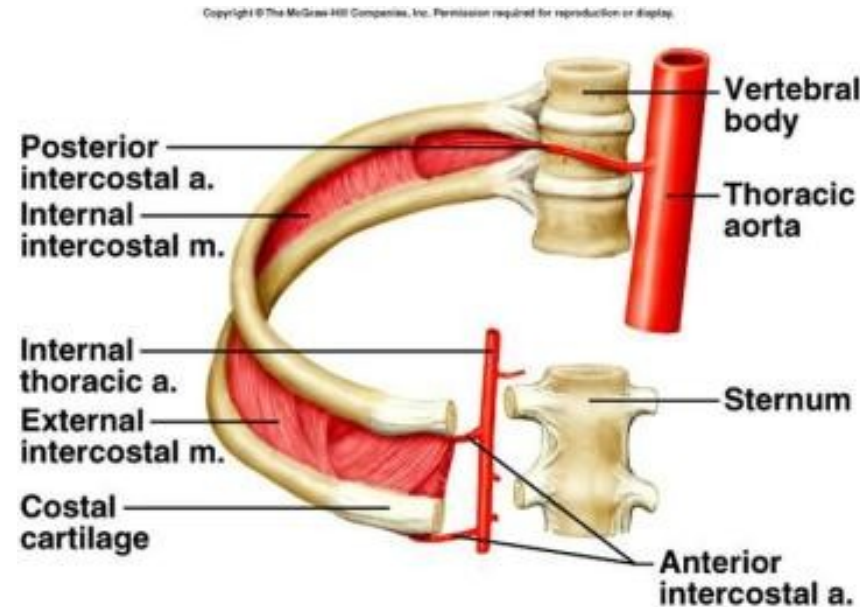
- The anterior ramus of the **1st thoracic nerve(T1)** first divides into a **superior** and a **inferior** part.
- The superior part joins the **brachial plexus** and the inferior part becomes the **1st intercostal nerve**.
- Other atypical features of specific intercostal nerves include the following:
 - The *1st and 2nd intercostal nerves* course on the **internal surface** of the 1st and 2nd ribs, instead of along the inferior margin in costal grooves
 - The *1st intercostal nerve* **has no anterior cutaneous** branch and often **no lateral cutaneous** branch.

ATYPICAL INTERCOSTAL NERVES con't ...

- The **2nd** *intercostal nerve* gives rise to a **large lateral cutaneous** branch (**intercostobrachial nerve**) which usually supplies the **floor of the axilla** and then communicates with the *medial brachial cutaneous nerve* to supply the **medial** and **posterior** surfaces of the **arm**.
- The **7th–11th** *intercostal nerves*, after giving rise to lateral cutaneous branches which continue to supply **abdominal** muscles and skin and then after they become **thoracoabdominal nerves** of the anterior abdominal wall.
- Their **anterior cutaneous** branches pierce the **rectus sheath**, becoming cutaneous close to the median plane.

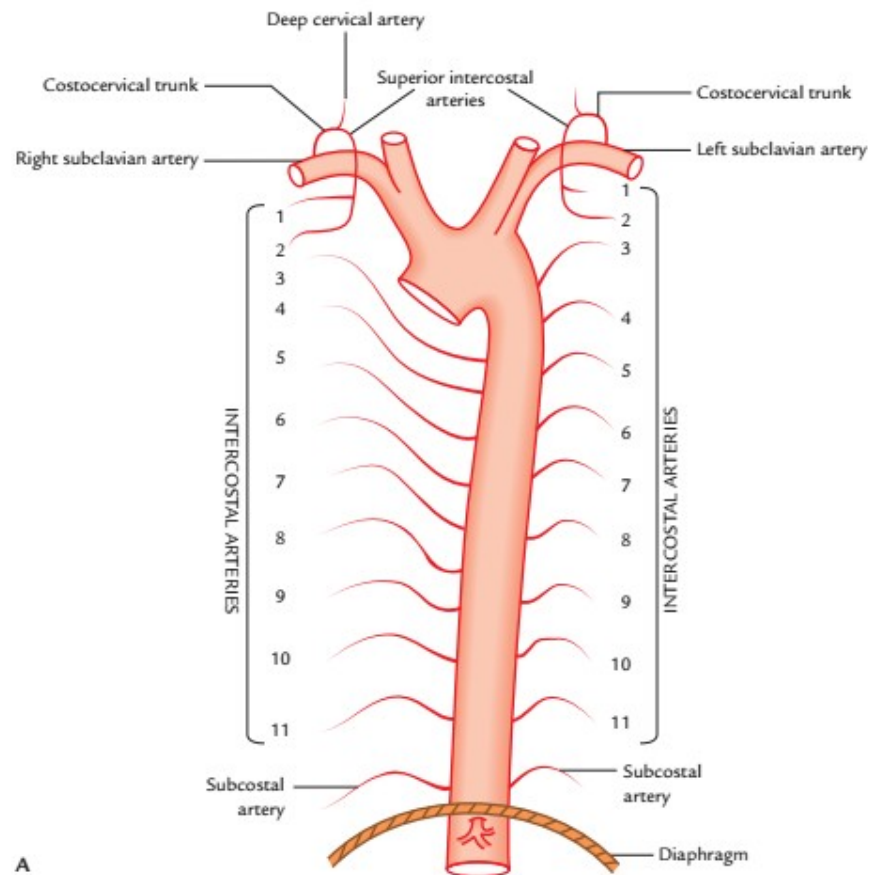
ARTERIES OF THORACIC WALL

- The arterial supply to the thoracic wall derives from the:
 - *Thoracic aorta*
 - *Subclavian artery*
 - *Axillary artery*
- The **intercostal arteries** course through the thoracic wall between the ribs.
- Each intercostal space is supplied by three arteries:
 - a large **posterior intercostal artery** (and its collateral branch)
 - a small **pair of anterior intercostal arteries**.



Posterior intercostal arteries

- 1st and 2nd posterior intercostal arteries arise from **costocervical trunk**
- 3rd -11th posterior intercostal arteries arise from **thoracic aorta** directly
- **Branches**
 - All give off a **dorsal branch** which supply the spinal cord, vertebral column, back muscles, and skin.
 - **collateral branch** that crosses the intercostal space and runs along the superior border of the rib.



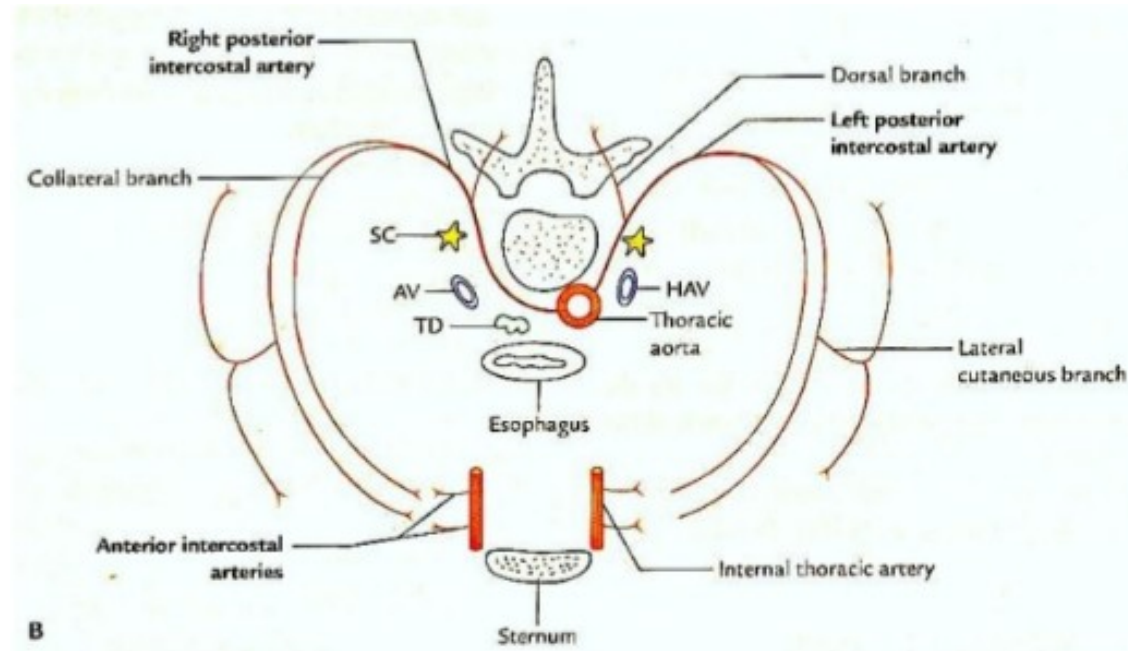
Branches of posterior intercostal A...

✓ **Dorsal branch**

✓ **Collateral branch**

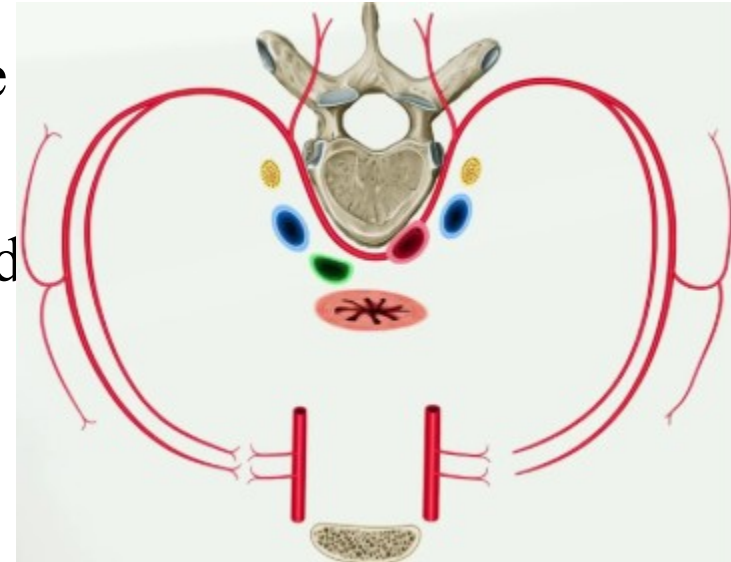
✓ **Muscular branch**

✓ **Lateral cutaneous branch**



Posterior intercostal arteries

- **Course of posterior intercostal arteries**
- At first the arteries run in the endothoracic fascia between the **parietal pleura** and the **internal intercostal membrane** then they run between the **innermost intercostal** and **internal intercostal** muscles.
- Close to the **angle** of the rib, the arteries enter the **costal grooves**, where they lie between the intercostal vein and nerve.
- Have terminal and collateral branches that **anastomose** anteriorly with **anterior intercostal arteries**



Anterior intercostal arteries

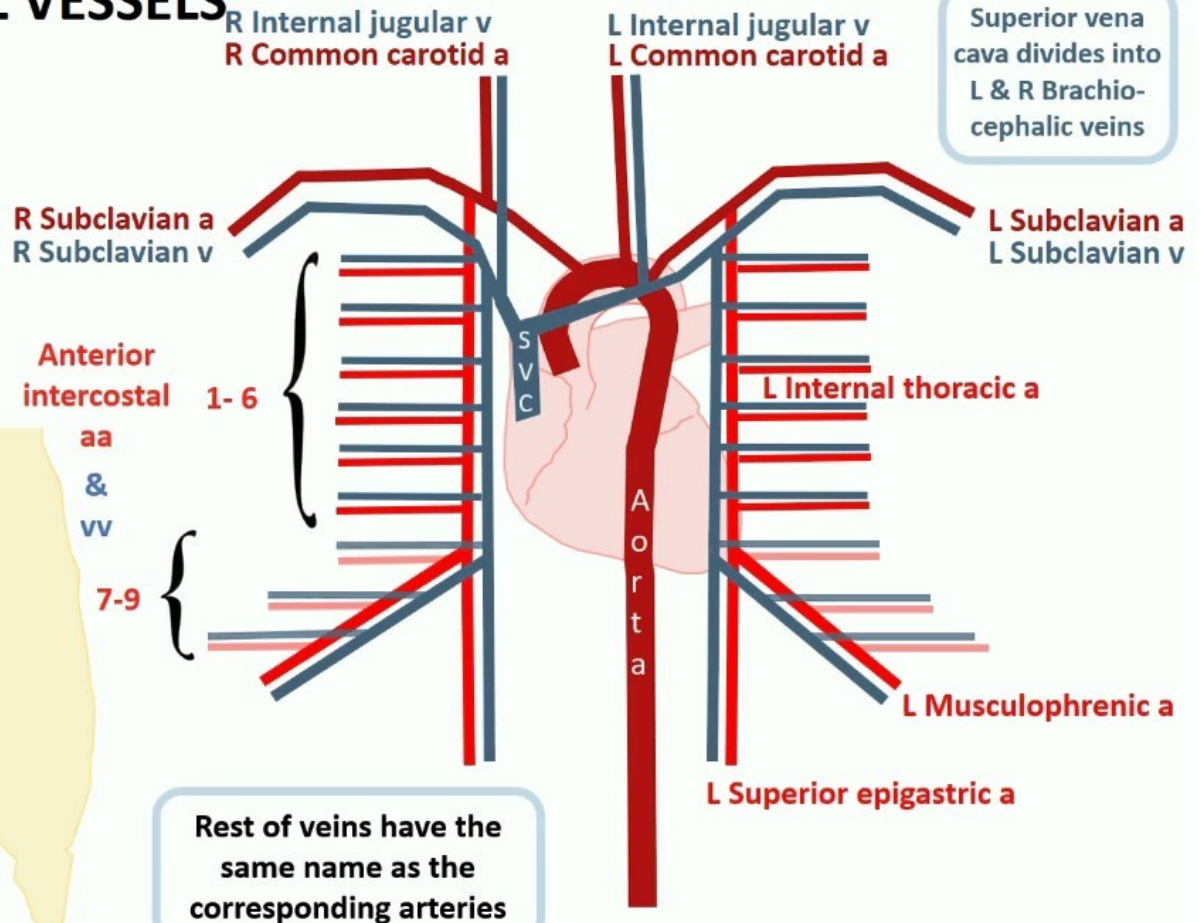
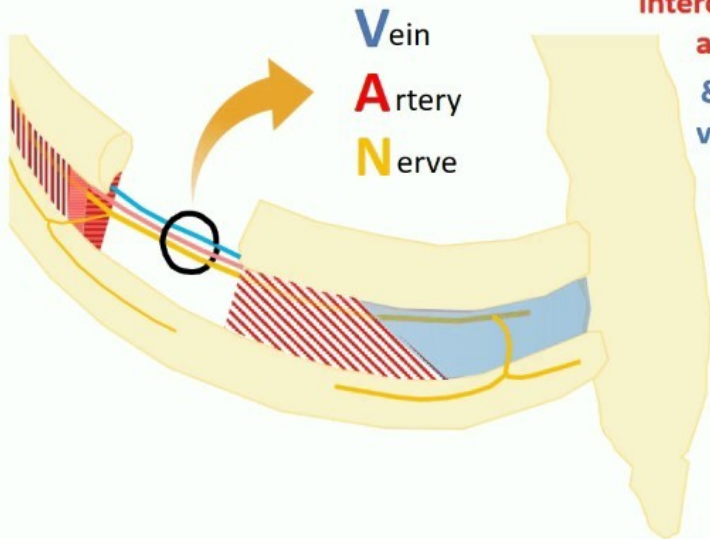
- Is a branch of **internal thoracic** artery
- Supply the anterior parts of the upper **9 intercostal** spaces.
- Are paired intercostal artery in each intercostal space
 - one near the inferior margin of the superior rib and
 - the other near the superior margin of the inferior rib.
- **1st – 6th** – from **internal thoracic (Mammillary artery)**
- **7th- 9th** – from **musculophrenic**
- **10th & 11th** – have **no anterior intercostal** ; these spaces are supplied only by the posterior intercostal arteries and their collateral branches.

Anterior intercostal arteries...

ANTERIOR INTERCOSTAL VESSELS

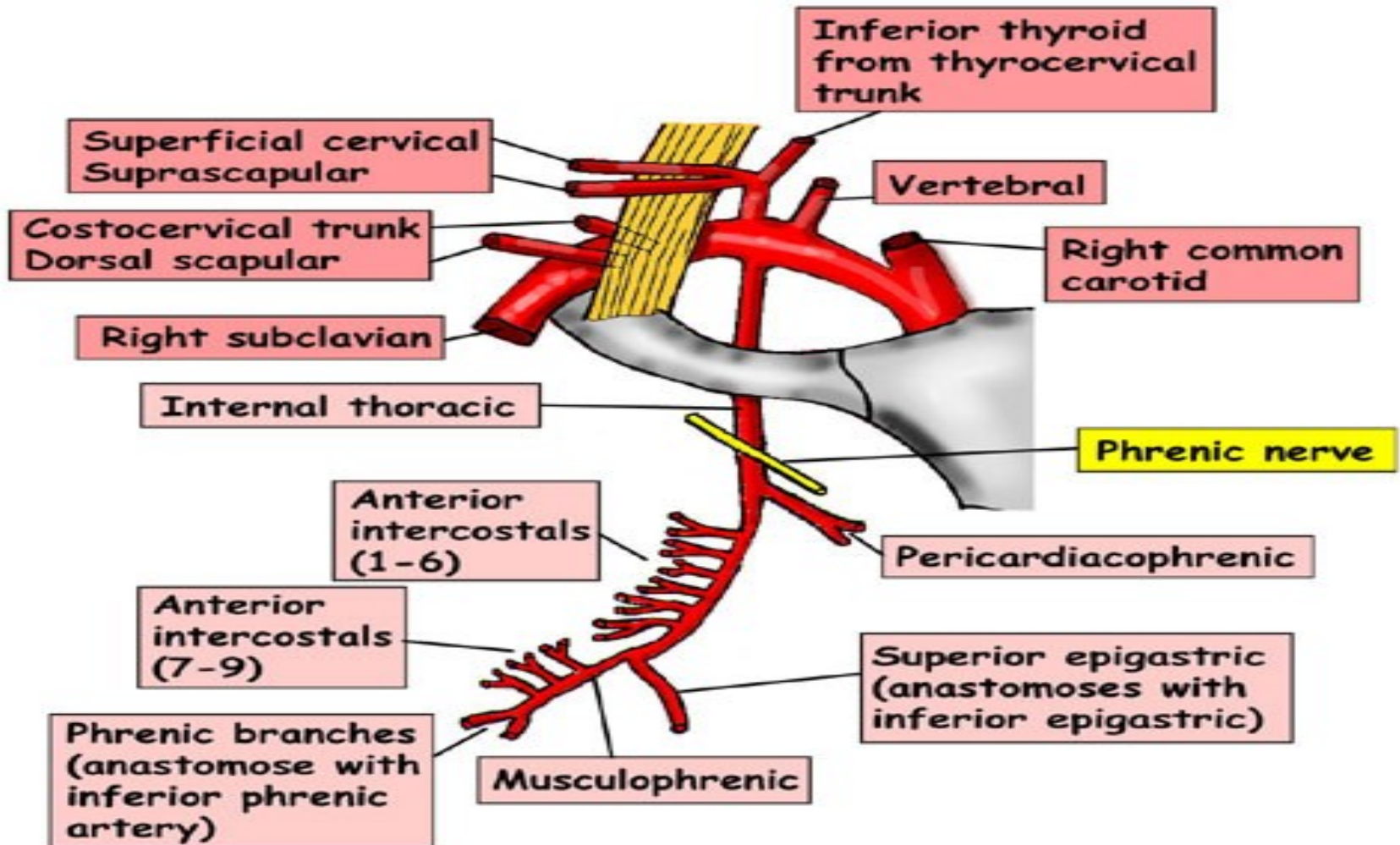
Know your ABCs

Aortic arch branches
Brachiocephalic trunk
Common carotid a (L)
Subclavian a (L)



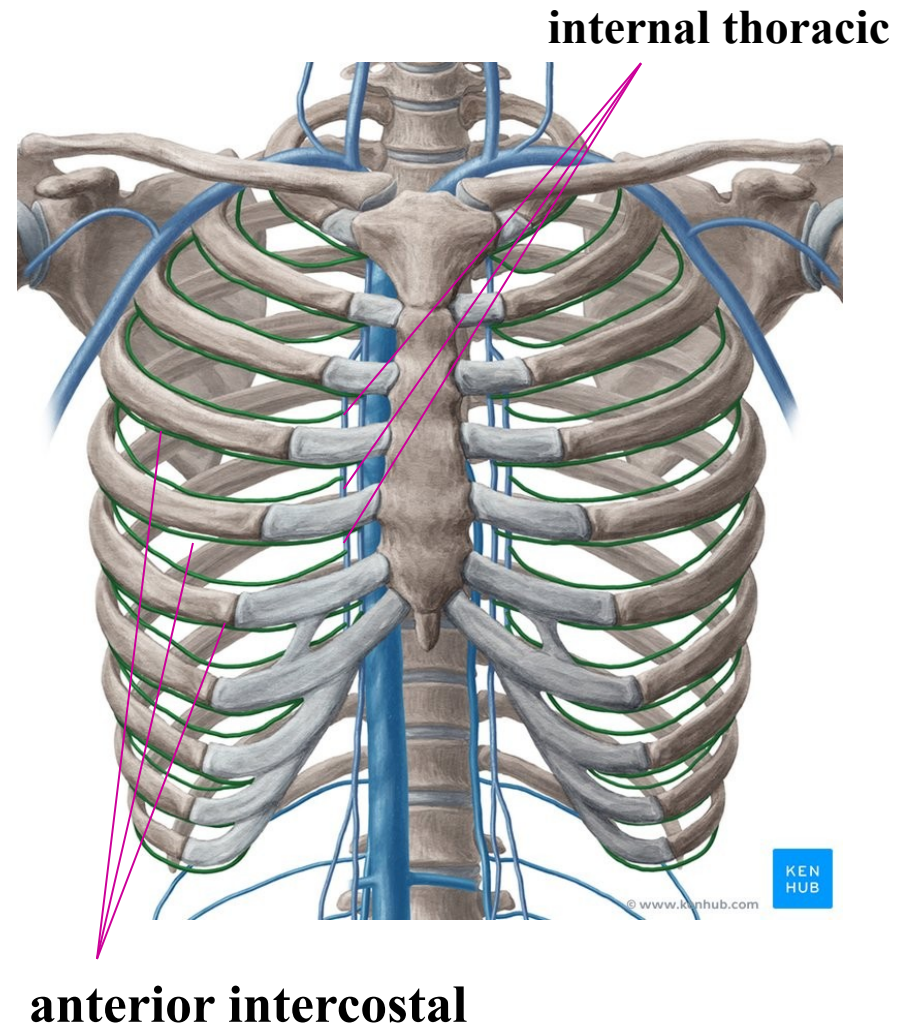
Vasculature of thoracic wall

RIGHT SUBCLAVIAN & INTERNAL THORACIC ARTERIES



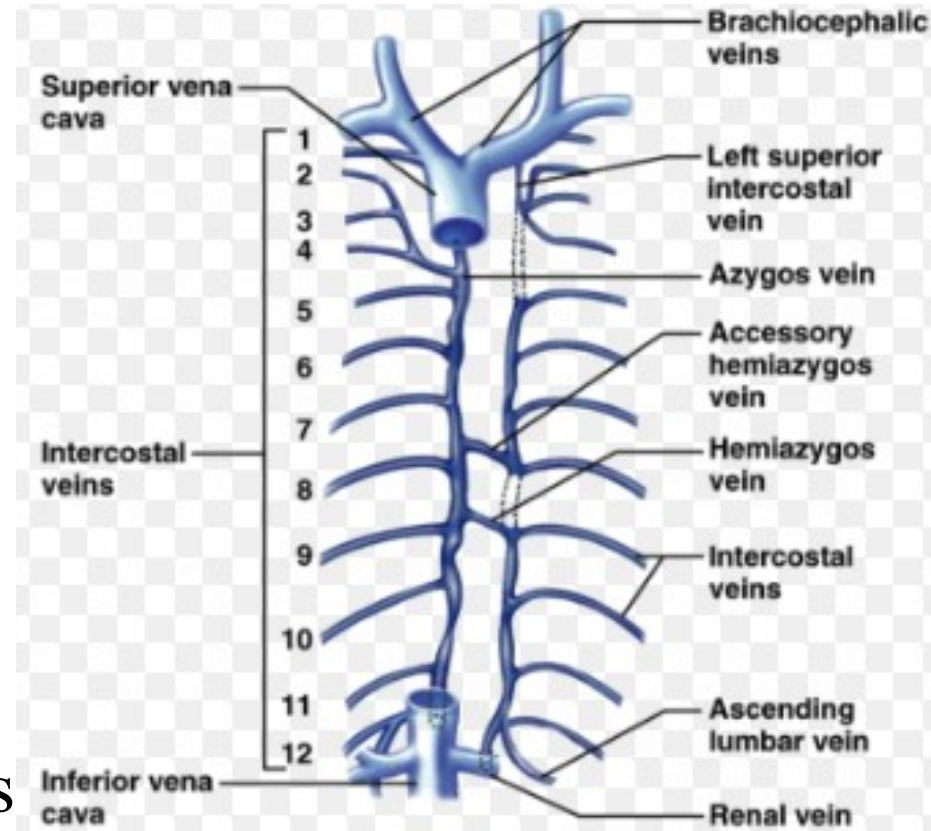
Intercostal veins

- The **intercostal veins** accompany the intercostal arteries and nerves and lie most superior in the costal grooves.
- **Contains**
 - Anterior intercostal veins
 - Posterior intercostal veins
- The **anterior intercostal** veins originate from the **intercostal** space just inferior to anterior aspects of their respective ribs and drain into the **internal thoracic** and **musculophrenic veins**.



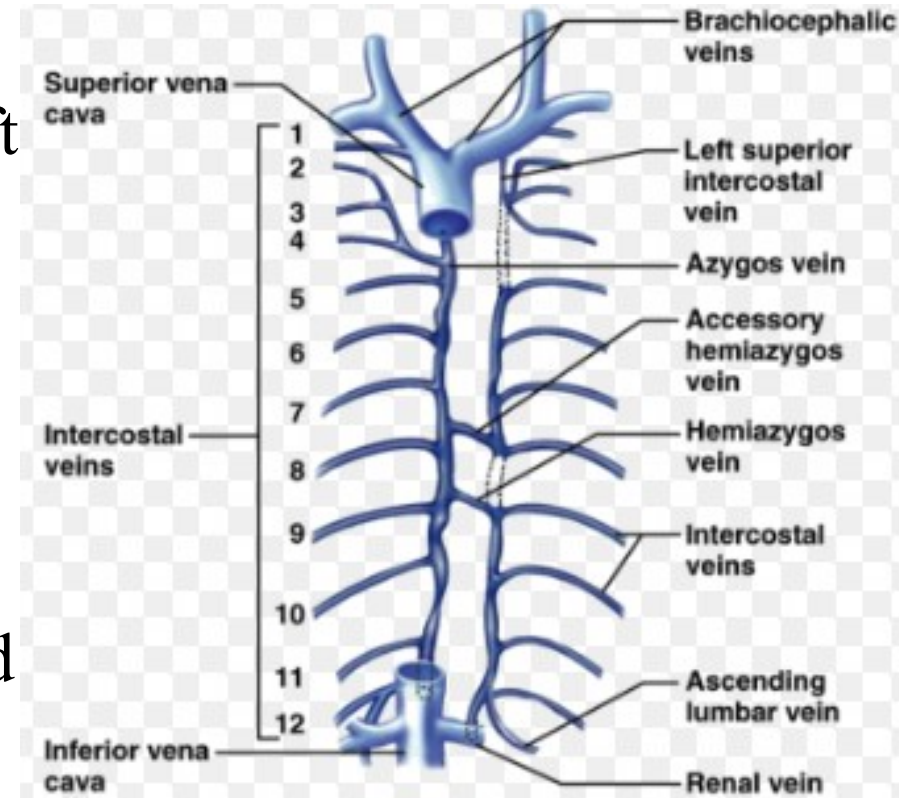
Posterior intercostal veins

- There are 11 **posterior intercostal veins** and one **subcostal vein** on each side.
- Most posterior intercostal veins drain into the **azygos (right side)** and **hemiazygos** and **accessory hemiazygos vein (left side)** which conveys venous blood to the superior vena cava (**SVC**).



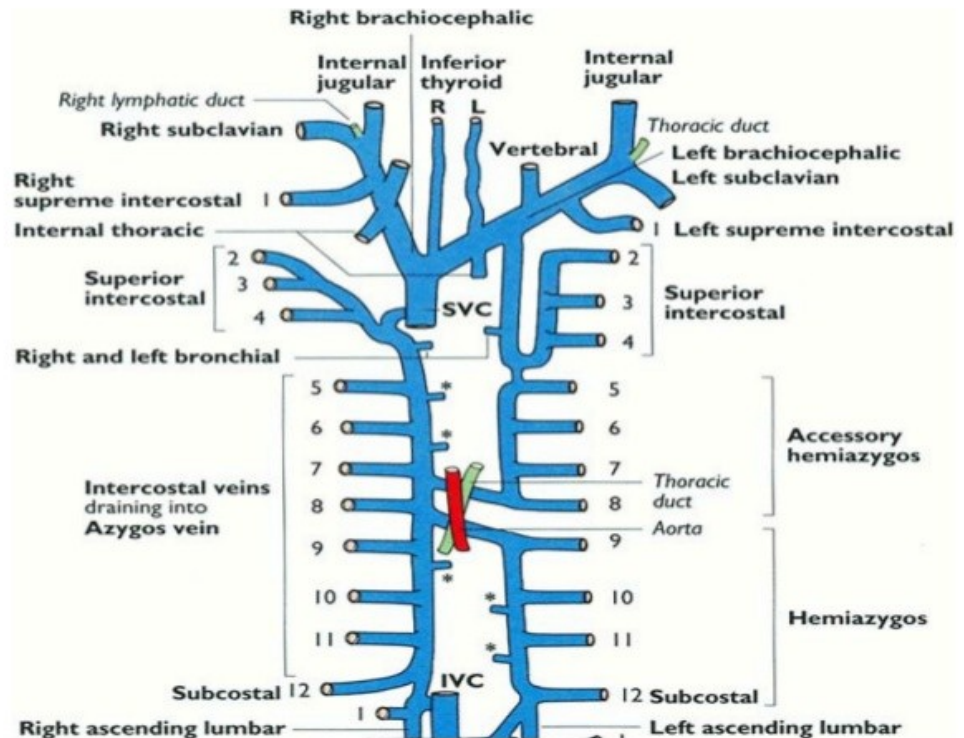
Posterior intercostal veins con't...

- Their **patterns are variable**, but they are commonly arranged as:
- The **1st** posterior intercostal vein, drains into the right and left **brachiocephalic vein**.
- The **2nd - 4th** drain into unite to form a trunk called the **superior intercostal vein (R&L)**
- The **right superior** intercostal vein draign into **Azygos** vein and the **left** one usually empties into the **left brachiocephalic vein**

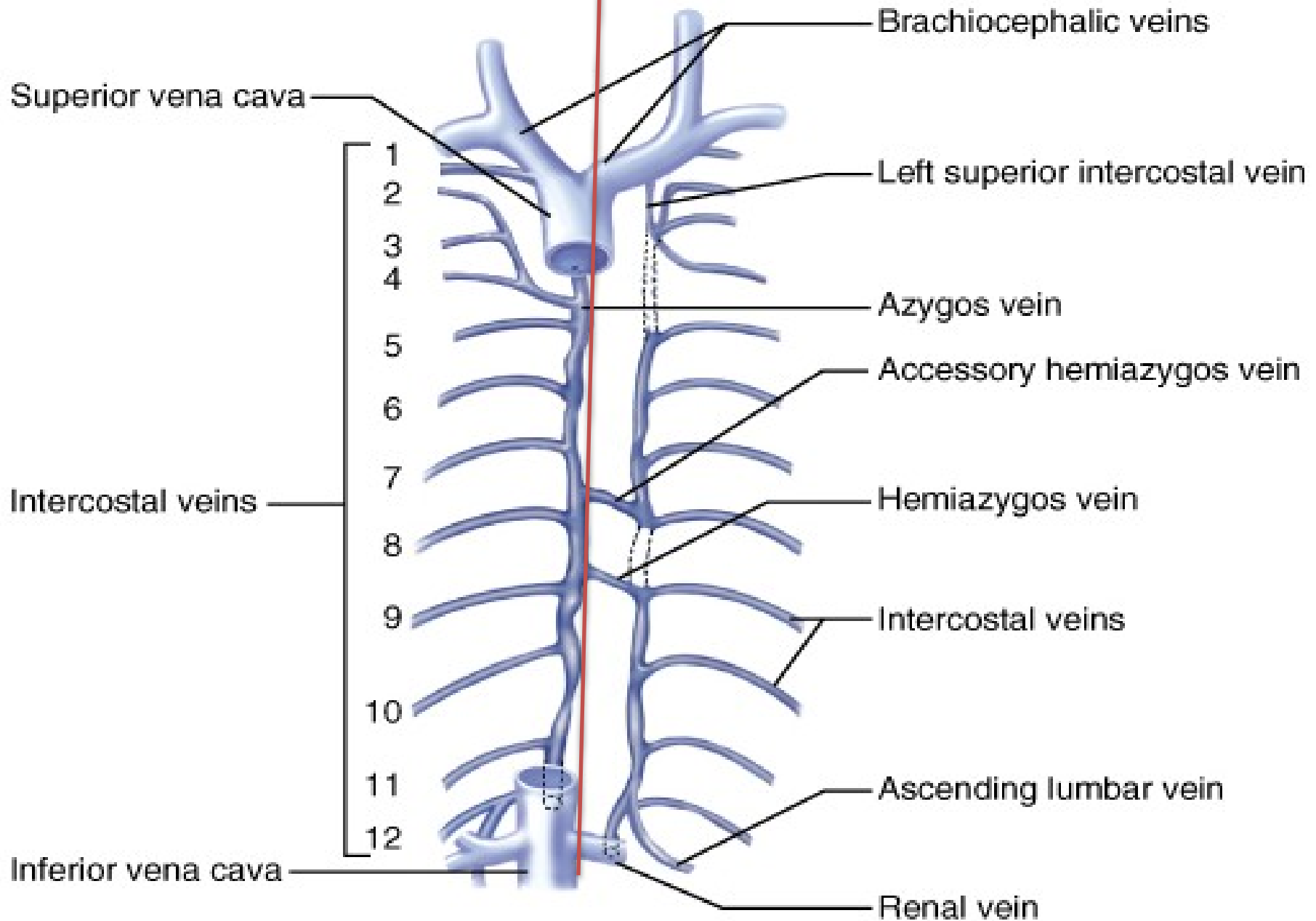


Posterior intercostal veins con't...

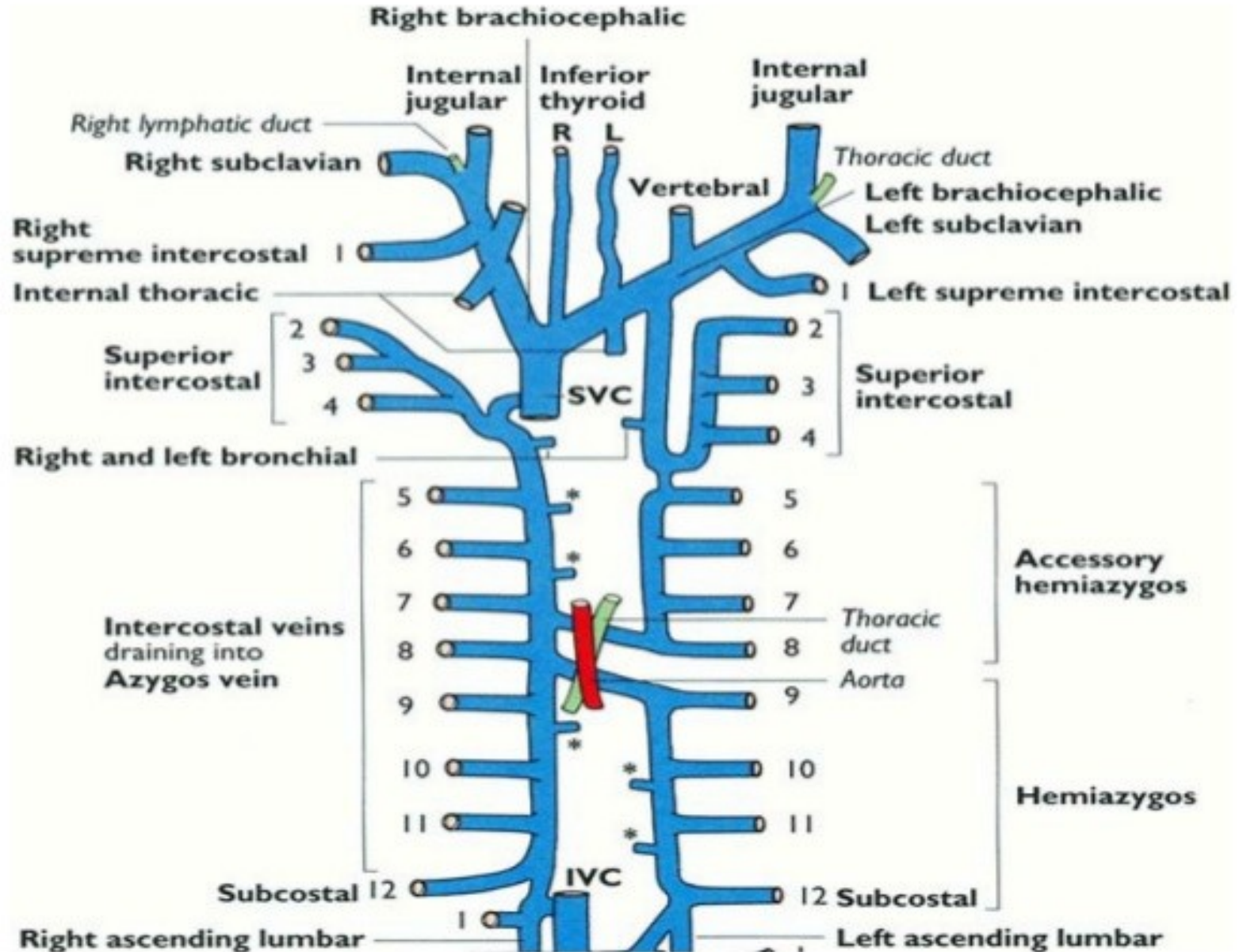
- The right & left 5th - 8th – drain into **azygos** & **accessory hemiazygos vein** respectively
- The right & left 9th -11th & subcostal – drain into **azygos hemiazygos vein** resp.



Intercostal veins



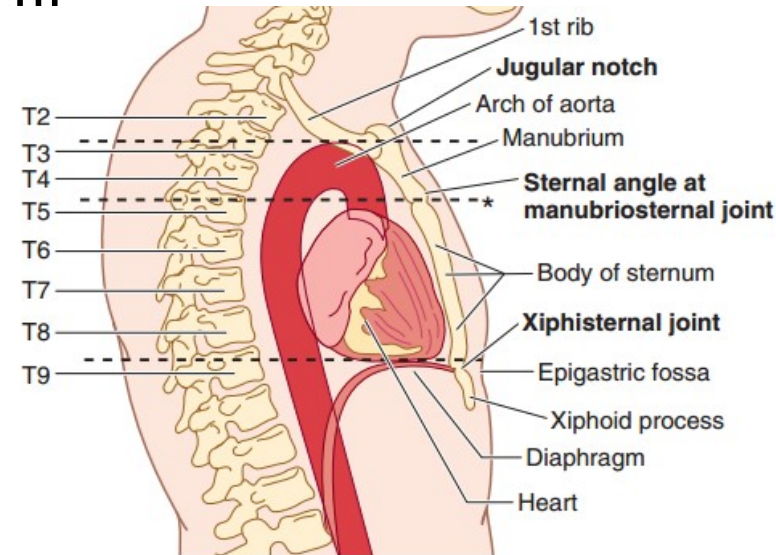
The internal thoracic veins...



Surface Anatomy of the Thorax

➤ The sternum

- The **jugular notch** in the manubrium can be palpated between the prominent medial ends of the clavicles.
- The notch lies at the level of the inferior border of the body of **T2** vertebra and the **space**.
- The **manubrium** lies at the level of **T3 & T4**.
- The **sternal angle**
 - lies at the level of the **T4–T5 IV disc**
 - It marks the level of the **2nd pair of costal cartilages**
 - left side of the manubrium is anterior to the **arch of the aorta** & **aortic arch**

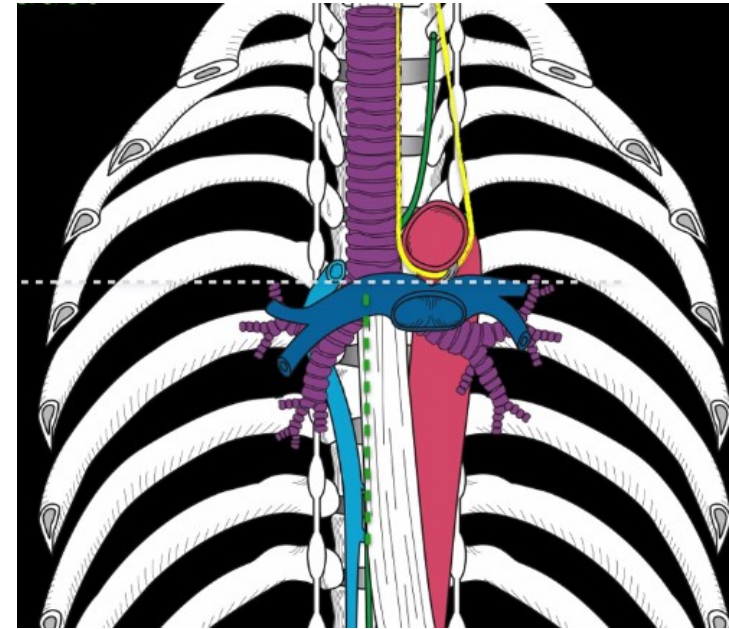


* Transverse thoracic plane

Surface Anatomy of the Thorax...

Sternal angle

- At the right side of manubrium **brachiocephalic** veins form the **superior vena cava (SVC)**.
- **Bifurcation of trachea** into the primary bronchi at this angle.
- **Bifurcation of pulmonary trunk**
- **Ligamentum arteriosum** arise at the level of TTP (Trance Thoracic Plane)
- **Nerves** such as vagus nerve and its branches
- **Thorasic duct** – at the level of sternal angle go to the **left side** of the vertebral body and ascend up then eventually to dumps in to left subclavian vein



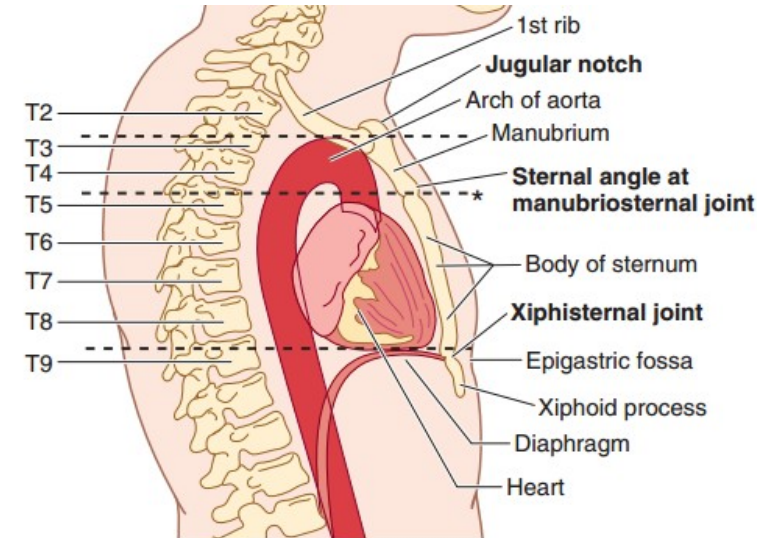
Surface Anatomy of the Thorax...

➤ The **body of the sternum**,

- Approximately 10 cm long
- lies anterior to the right border of the heart and vertebrae T5–T9.
- The **intermammary cleft** overlies the sternal body.

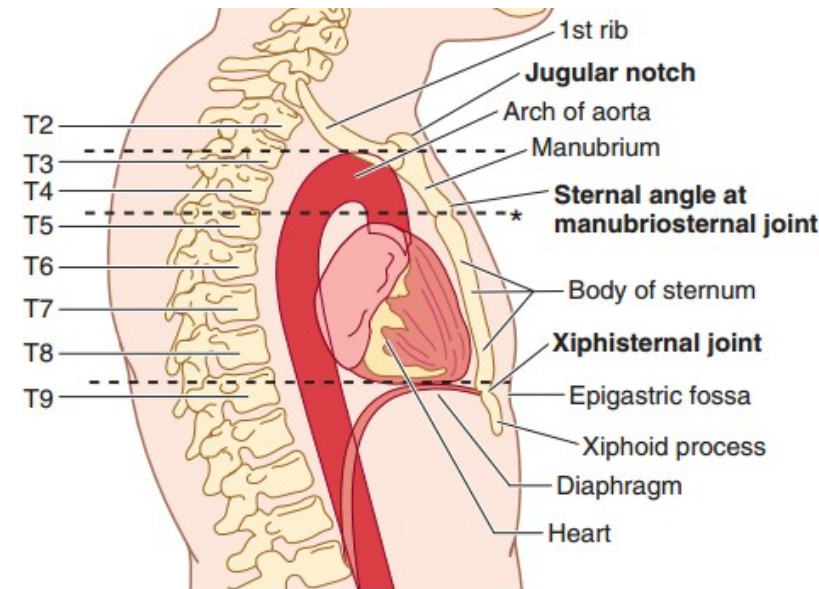
➤ The **xiphoid process**

- lies in a slight depression, the **epigastric fossa**.
- lies at the level of T10 vertebra.
- The **costal margins**, formed by the joined costal cartilages of the 7th–10th ribs, are easily palpable



Surface Anatomy of the Thorax...

- The **xiphoid process** is an important **landmark** in the median plan
 - at xiphisternal joint (at level of inferior border of **T9 vertebra**) indicates the **inferior limit** of central part of **thoracic cavity**.
 - A midline marker for **superior limit of liver, central tendon of diaphragm & inferior border of the heart**.



Surface Anatomy of the Thorax...

Imaginary Lines: facilitate anatomical & clinical description of the body

- The following lines are extrapolated over the thoracic wall based on visible or palpable superficial features:-

Anterior median (midsternal) line (AML)

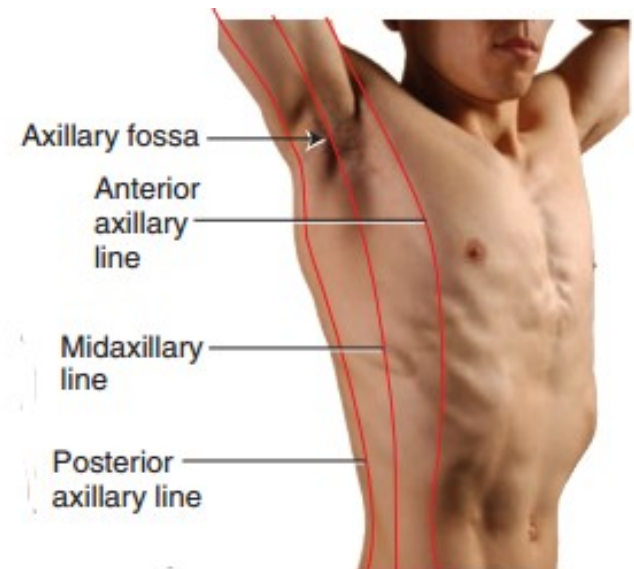
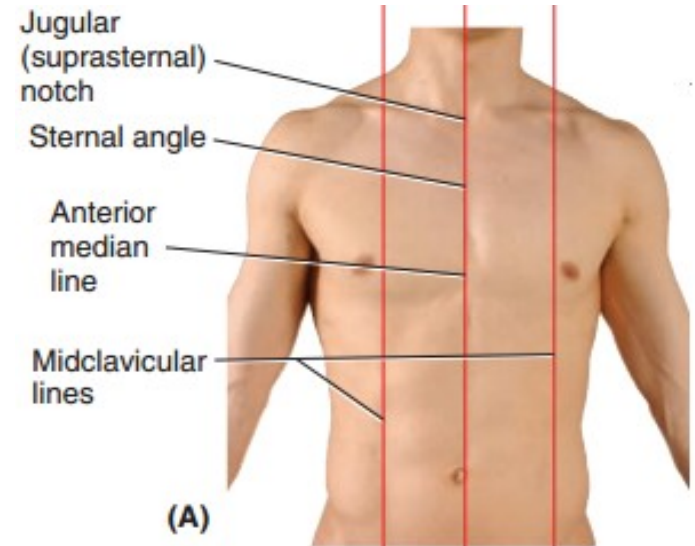
- Indicates intersection of the median plane with the anterior thoracic wall.

Mid clavicular line (MCL)

- passes thru mid point of clavicle, parallel to the AML.

Anterior axillary line (AAL)

- runs vertically along anterior axillary fold that is formed by inferolateral border of pectoralis major.



Surface Anatomy of the Thorax...

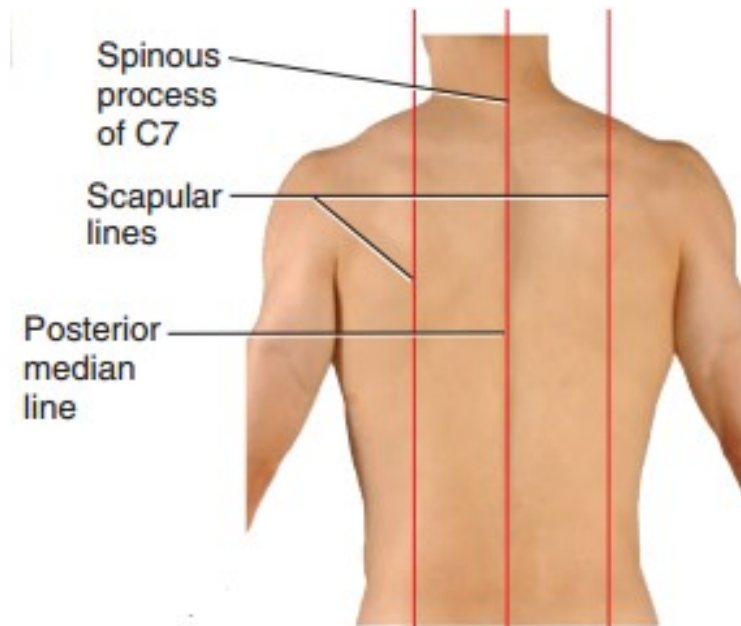
Mid axillary line (MAL) –

- runs from apex (deepest part) of axillary fossa, parallel to AAL

Posterior axillary line- runs from posterior axillary fold that is formed by *latissimus dorsi & teres major*, which is parallel to MAL

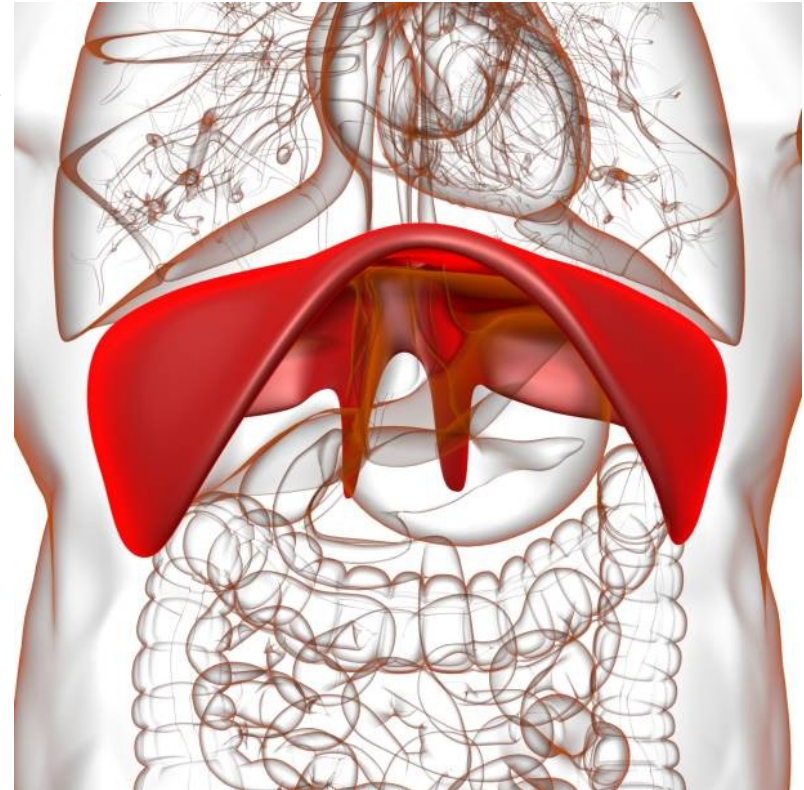
Posterior median line (PML) –a vertical line along tips of spinous process of the vertebrae.

Scapular line (SL) – is parallel to the posterior median line & intersects inferior angles of scapula.



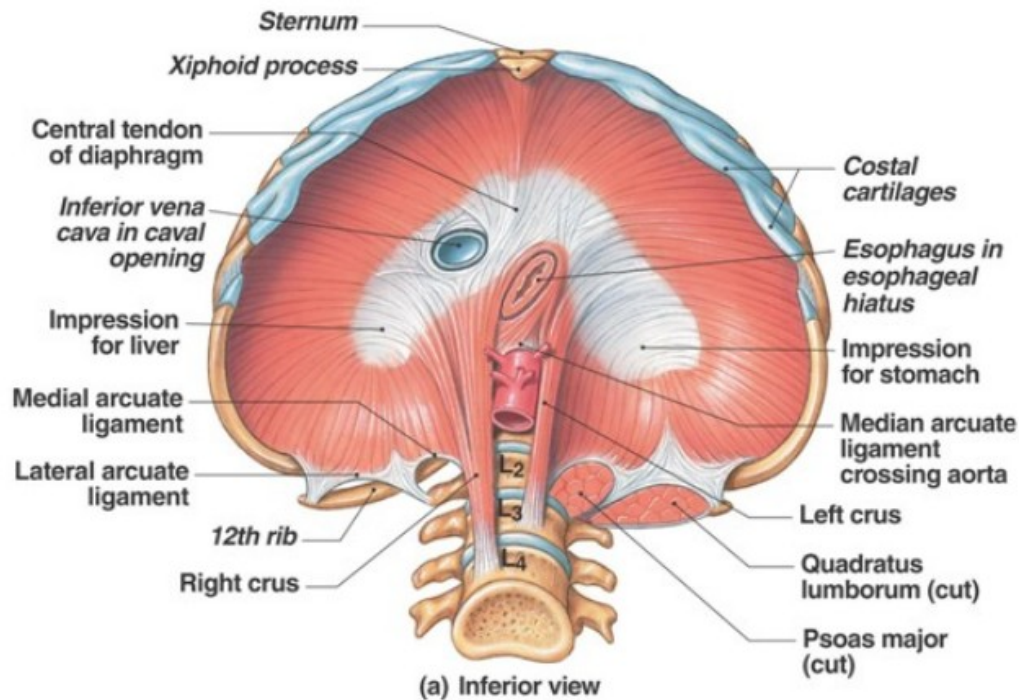
Diaphragm

- Dome-shaped musculotendinous partition between **thoracic** and **abdominal** cavities
- Principal muscle of respiration
- Composed of two portions:
- Muscular (peripheral part)
- Aponeurotic (central part)



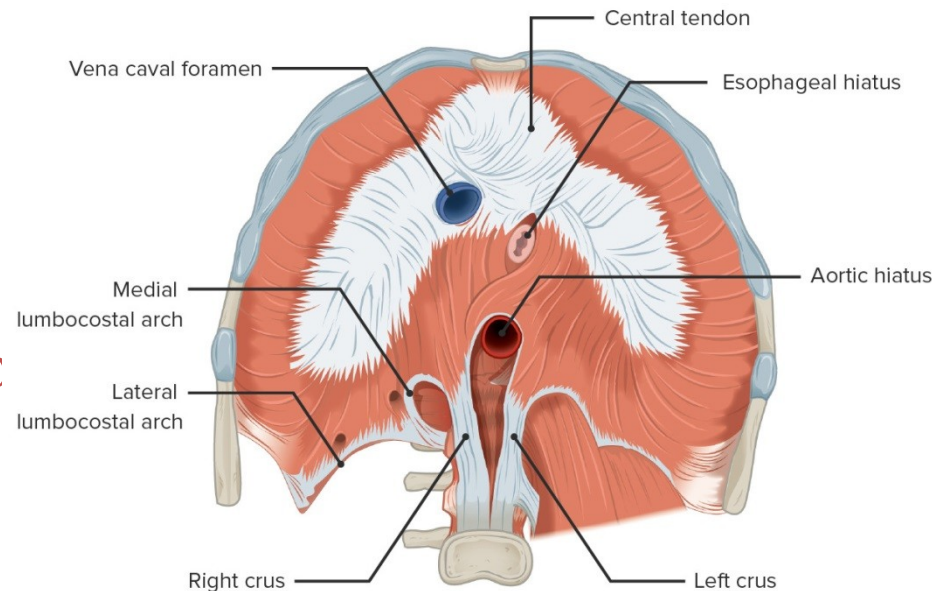
Diaphragm...

- Muscular part
- Fibers converge radially to central tendon
- divided into 3 parts based on based on the peripheral attachments:
 - **Sternal part** – that attach to the posterior aspect of the xiphoid process
 - **Costal part** – inner surface of lower six costal cartilage
 - **Vertebral (Lumbar) part** – lumbar vertebrae by two crura



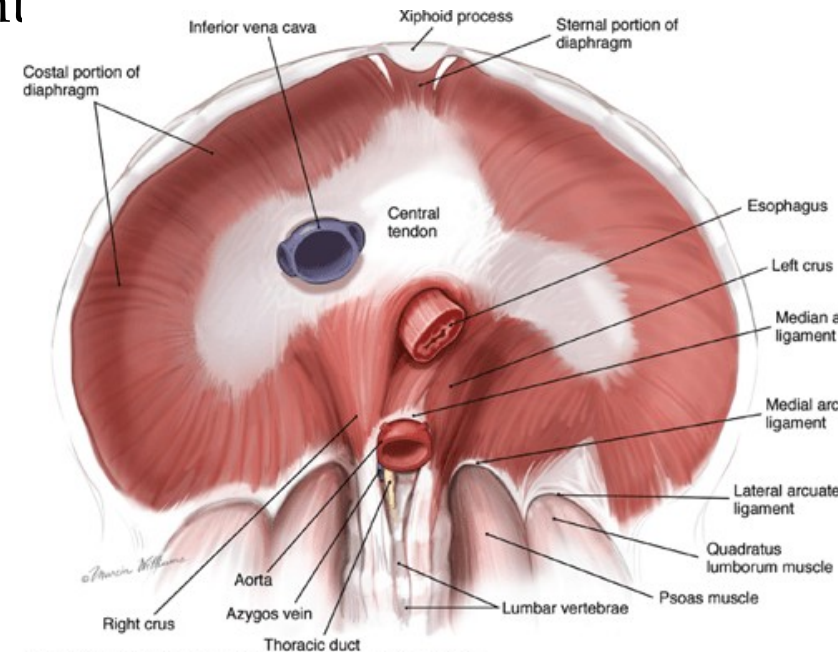
Diaphragm

- **Right Crus** –larger and longer than the left crus, arises from **superior 3 or 4** lumbar vertebrae.
- **Left Crus** – arises from **superior 2 or 3** lumbar vertebrae.
- Crura are united superiorly at T12 by narrow arch called **median arcuate ligament** which form the **aortic hiatus**.



Structures passing through diaphragm

- **Vena caval opening-** T8 right of the median plane with in the central tendon, transmits IVC, branch of right phrenic nerve, lymph vessels from liver.
- **Esophageal opening-** T10 left of the median plane, transmits, esophagus, vagus nerves, left gastric vessels, lymphatic vessels
- **Aortic** – T12 median plane, transmits azygos vein, hemi-azygos vein, thoracic duct



Source: D. J. Sugarbaker, R. Bueno, Y. L. Colson, M. T. Jakitsch, M. J. Krasna, S. J. Mentzer, M. Williams, A. Adams: Adult Chest Surgery, 2nd Edition: www.accesssurgery.com Copyright © McGraw-Hill Education. All rights reserved.

Vessels and nerves of Diaphragm

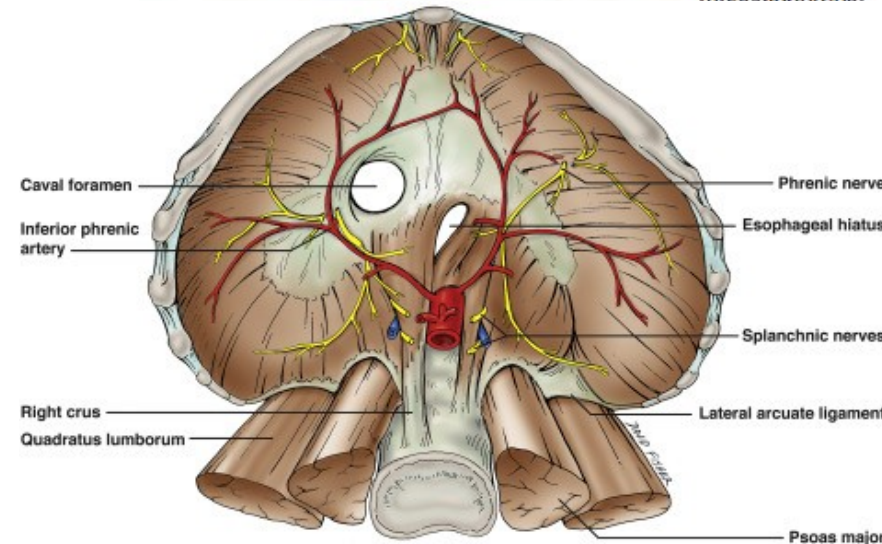
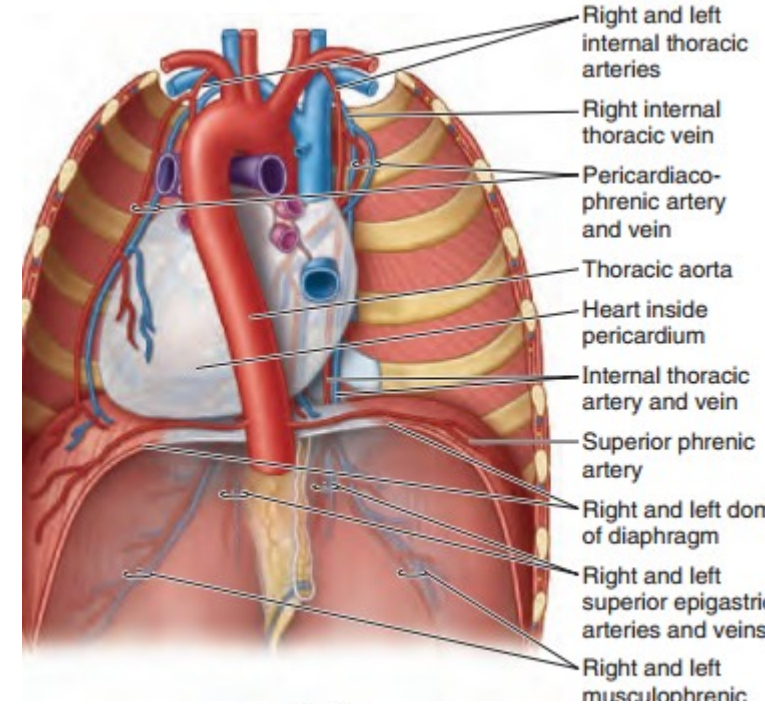
Arteries

- ❖ Superior surface
 - ✓ Superior phrenic (thoracic aorta)
 - ✓ **Musculophrenic** and **pericardiophrenic** (internal thoracic artery)
- ❖ Inferior surface: **inferior phrenic**

Veins: veins accompany artery with similar name

Nerve supply

- Phrenic (motor and sensory)
- Lower six intercostal and subcostal for peripheral part (sensory)



Thank you!